ArbRPN: A Bidirectional Recurrent Pansharpening Network for Multispectral Images With Arbitrary Numbers of Bands

Lihui Chen, Zhibing Lai, Gemine Vivone, *Senior Member, IEEE*, Gwanggil Jeon, *Senior Member, IEEE*, Jocelyn Chanussot, *Fellow, IEEE*, and Xiaomin Yang, *Member, IEEE*

***Abstract*— Although the performance of pansharpening has been significantly improved by advanced deep-learning (DL) technologies in recent years, most DL-based methods fail to process multispectral (MS) images with arbitrary numbers of bands by a single model. Consequently, it is inevitable to train separate models for MS images with different numbers of bands, which is time- and storage-consuming as well as inefficient in practice. To tackle the above problem, we propose a bidirectional recurrent pansharpening network (named ArbRPN) for MS images with arbitrary numbers of bands. Our ArbRPN can dynamically reconstruct high-resolution (HR) MS images with different numbers of bands by adaptively changing the number of recurrence to the number of bands of the low-resolution (LR) MS images. Leveraging on the ability of the ArbRPN to process MS images with any number of bands, one can even customize the bands to be pansharpened. Moreover, to achieve superior performance, spectral discrepancy and dependence are considered in the ArbRPN. Details from the panchromatic (PAN) image are adaptively injected into the fused product according to the captured spectral dependence. Furthermore, training strategies of existing DL-based pansharpening methods can only group MS images with a constant number of bands into mini-batches. Therefore, we present a mask-based training method (called mask-training) to solve this problem. Benefiting from the mask-training, our ArbRPN can achieve superior performance and robustness during pansharpening. Extensive experiments show the superior performance of our ArbRPN with respect to the state-of-the-art (SOTA) methods applied to MS images with different numbers of bands. The code of our ArbRPN is available on https://github.com/Lihui-Chen/ArbRPN.git.**

Manuscript received August 29, 2021; revised November 1, 2021; accepted November 20, 2021. Date of publication November 29, 2021; date of current version February 21, 2022. This work was supported in part by the National Key Research and Development Program of China under Grant 2020AAA0104500, in part by Sichuan University under Grant 2020SCUNG205, and in part by the China Scholarship Council (CSC) under Grant 202006240191. *(Corresponding author: Xiaomin Yang.)*

Lihui Chen, Zhibing Lai, and Xiaomin Yang are with the College of Electronics and Information Engineering, Sichuan University, Chengdu 610065, China (e-mail: lihuichen@stu.scu.edu.cn; laizhibing@stu.scu.edu.cn; arielyang@scu.edu.cn).

Gemine Vivone is with the National Research Council, Institute of Methodologies for Environmental Analysis (CNR-IMAA), 85050 Tito Scalo, Italy (e-mail: gemine.vivone@imaa.cnr.it).

Gwanggil Jeon is with the Department of Embedded Systems Engineering, Incheon National University, Incheon 22012, South Korea (e-mail: ggjeon@gmail.com).

Jocelyn Chanussot is with the Laboratoire Jean Kuntzmann (LJK), Université Grenoble Alpes, Inria, CNRS, Grenoble INP, 38000 Grenoble, France, and also with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China (e-mail: jocelyn.chanussot@grenoble-inp.fr).

Digital Object Identifier 10.1109/TGRS.2021.3131228

***Index Terms*— Deep learning (DL), image fusion, multispectral (MS) images, pansharpening, recurrent neural networks (RNNs), remote sensing.**

I. Introduction

DUE to the limitation of imaging devices, remote sensing satellites cannot obtain an image with both high spatial and high spectral resolution. Therefore, satellites alternatively capture a low-resolution multispectral (LRMS) image containing rich spectral information and a high-resolution panchromatic (PAN) image containing fine spatial information. However, high-resolution multispectral (HRMS) images are always preferred in subsequent applications, since HRMS images facilitate perception and decision making for humans or machines. To obtain HRMS images, pansharpening is proposed to fuse the complementary information from the LRMS and PAN images.

Benefiting from the development of deep learning (DL), the performance of pansharpening has been significantly promoted in recent years. However, most DL-based methods require separate models to process MS images with different numbers of bands leading to two main issues. First, the separate training of models for MS images with different numbers of bands always costs lots of computational resources and training time. Second, the storing of numerous distinct models takes considerable storage resources. Therefore, it is inefficient and impractical to train numerous networks for MS images with a large range of the number of bands. Moreover, it is inevitable to encounter MS images with different numbers of bands when one attempts to process MS images from different sensors by a single model meaning that a unified DL-based model for MS images from arbitrary sensors is a crucial problem to solve. Although a few DL-based methods [2], [3] can process LRMS images with different numbers of bands by reconstructing each band independently (as detailed in Section II), they neglect the spectral discrepancy and dependency among different bands in MS images resulting in poor performance from a spectral preservation point of view. As shown in Fig. 1(a), the relative spectral response curves of the QuickBird (QB) bands exhibit the spectral discrepancy among different bands, as the peak of each band locates at different wavelengths. Instead, a spectral overlap exists not only among adjacent MS bands but also between each band and the PAN image demonstrating that

1558-0644 © 2021 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 12:30:08 UTC from IEEE Xplore. Restrictions apply.

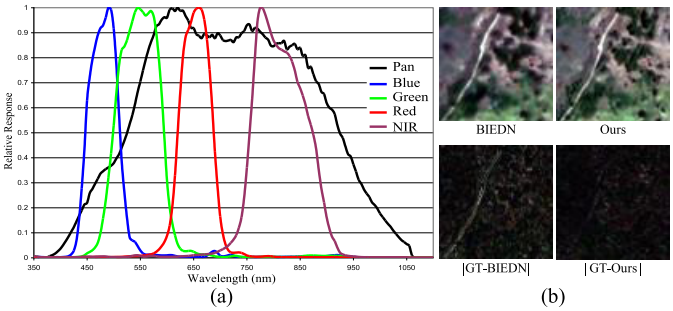


Fig. 1. (a) Relative spectral response curves of the QB bands (see [1]). (b) (Top row) Results obtained by [2] and by the proposed ArbRPN; (bottom row) residual images obtained by subtracting the ground-truth (GT) and the outcomes obtained by [2] and ArbRPN, respectively.

the spectral dependency is a typical feature for MS and PAN images. Besides, the reflection spectrum of a target always covers a large range leading to similar structures in various spectral bands. This factor also remarks the spectral dependency among bands. Since the spectral discrepancy and dependency are neglected in [3] and [2], the results obtained by the above methods often encounter issues as a remarkable spectral distortion.

To address the above issues, we propose a bidirectional recurrent pansharpening network, named ArbRPN, for MS images with arbitrary numbers of bands, in which the spectral discrepancy and dependency are considered. First, to process MS images with any number of bands, the recurrence mechanism of recurrent neural networks (RNNs) [4] is utilized in our ArbRPN. By this recurrent mechanism, our ArbRPN can dynamically reconstruct HRMS images with any number of bands via adaptively changing the number of recurrences. Second, to capture the spectral dependency and to consider the spectral discrepancy, a sequential bidirectional RNN (BiRNN)-based architecture is devised in our ArbRPN based on three properties (see Section III-A). Based on the captured spectral dependency, the proposed ArbRPN can adaptively inject band-dependent details into interpolated LRMS images. As shown in Fig. 1(b), having considered the spectral discrepancy and dependency, the proposed network achieves better results than [2]. Moreover, for superior performance and robustness, a mask-based training method called mask-training is proposed to train the proposed ArbRPN using MS images with arbitrary numbers of bands.

To sum up, the contributions of the article are as follows.

1) We propose a bidirectional recurrent network (named ArbRPN) to pansharpen MS images with arbitrary numbers of bands by a single model. Unlike prior networks, which process each band independently, to the best of our knowledge, this is the first pansharpening network that takes the spectral discrepancy and dependency into consideration to address the fusion of MS images with arbitrary numbers of bands. Leveraging this latter ability, one can even customize the bands to be pansharpened.
2) We present a sequential BiRNN-based structure to capture the spectral dependency and inject the details from PAN images according to captured spectral dependency. Compared with the vanilla BiRNN, the sequential BiRNN shows to be more stable during the training in our experiments. Besides, using a similar number of parameters and by stacking convolutional layers along the network depth direction in the sequential BiRNN, a deeper network than the vanilla BiRNN (where the convolutional layers are parallel) can be obtained. These characteristics allow the proposed ArbRPN to get better performance.
3) We propose a mask-based training strategy to train ArbRPN using MS images with arbitrary numbers of bands, thus facilitating the performance and the robustness during processing MS images with different numbers of bands. To the best of our knowledge, this is the first attempt to train a pansharpening network with mini-batches consisting of MS images with different numbers of bands.
4) We also introduce three simple yet effective properties in Section III-A for spectral-dependent pansharpening networks, which derive from the nature of the spectral discrepancy and dependency among various bands in MS and PAN images. These properties can inspire some subsequent spectral-dependent pansharpening networks.

II. Related Works

A. Traditional Pansharpening Methods

Traditional pansharpening methods can be classified into three groups: component substitution (CS), multiresolution analysis (MRA), and optimization-based methods. In CS-based methods, the MS image is decomposed into the spatial and the spectral component, and then the spatial component is replaced by the PAN image. Haydn [5] and Carper *et al.* [6] used intensity-hue-saturation (IHS) transformation to decompose the MS image replacing the spatial component with the PAN image. Gillespie *et al.* [7] presented the Brovey transformation for pansharpening. Laben and Brower [8] introduced Gram–Schmidt (GS) transformation for pansharpening. Rahmani *et al.* [9] improved the IHS and presented the adaptive IHS coefficients to obtain more accurate spectral information. To process data from various types of sensors, Choi *et al.* [10] proposed an adaptive CS-based methods by partial replacement. Xu *et al.* [11] adopted a data fitting method in well-established CS-based methods to alleviate the spectral distortion of CS-based methods. Garzelli *et al.* [72] proposed the band-dependent spatial details (BDSD) injection model. Vivone *et al.* [12], [13] introduced the BDSD robust version adding physical constraints (BDSD-PC) to overcome the limited performance of BDSD during processing MS images with more than four bands. Based on the GS [8], Aiazzi *et al.* [14] proposed a GS adaptive (GSA) method. Although CS-based methods were efficient, these methods always resulted in spectral distortions due to inappropriate decomposition solutions. Therefore, MRA-based methods emerged to alleviate this drawback by injecting the details extracted from PAN images by multiscale decomposition into the interpolated LRMS image. As MRA-based methods rarely process the spectral information in the LRMS, results obtained by them generally

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 12:30:08 UTC from IEEE Xplore. Restrictions apply.

have better spectral preservation than CS-based methods. Some research studies for MRA-based methods are focused on different methods for multiscale decomposition, such as generalized Laplacian pyramid (GLP) [15], wavelet transformation [16], and contourlet transformation [17]. In MRA-based methods, results often contain spatial artifacts due to an inappropriate multiscale decomposition. Under mild hypotheses, Tu *et al.* [18] integrated the CS- and MRA-based methods into a details injection framework, which was formulated as follows:

$$\widehat{\mathbf{HRMS}}_i = \widetilde{\mathbf{MS}}_i + g_i(\mathbf{Pan} - \mathbf{I}_L) \tag{1}$$

where $\widehat{\mathbf{HRMS}}_i$ and $\widetilde{\mathbf{MS}}_i$ represent the ith band of the HRMS image and the ith interpolated LRMS image, respectively, **Pan** is the PAN image, $\mathbf{I}_L$ is the low-frequency component obtained from the MS image or the PAN image, and g_i is the injection coefficient for the ith band. As for CS-based methods, $\mathbf{I}_L$ is function of the MS spectral bands. Conversely, if $\mathbf{I}_L$ is obtained from the PAN image, (1) represents the fusion equation of an MRA-based method. In [19], high pass modulation (HPM)-based and error-based injection models are proposed. To improve the performance, a regression-based approach was introduced in [20] to estimate the injection coefficients. Further improving the performance, Restaino *et al.* [21] segmented images to estimate context-adaptive injection coefficients. Moreover, a full scale (FS) injection model based on regression was introduced in [22]. In [23], the framework of the robust regression is applied to improve the injection coefficients of the context-based GLP for pansharpening. Finally, a haze correction was shown to be beneficial to MRA-based pansharpening in [24].

To achieve a tradeoff between spatial and spectral distortions, some variational optimized-based methods are presented. Ballester *et al.* [25] proposed the pioneer variational method for PAN and MS image fusion. Fang *et al.* [26] introduced a new variational method for pansharpening based on three assumptions, thus building an energy functional, whose minimizer is related to the best pansharpened result. Recently, Deng *et al.* [27] proposed a reproducing kernel Hilbert space and approximated Heaviside function-based variational method for pansharpening. Liu *et al.* [28] presented a variational method based on spatial fractional order geometry and spectral low-rank prior. Fu *et al.* [29] introduced local gradient constraints for variational pansharpening. From a Bayesian point of view, Mascarenhas *et al.* [30] presented first a Bayesian-based pansharpening. Fasbender *et al.* [31] pointed out that the weights of the spectral and the spatial information can be adaptable by weighting criteria according to users' needs. Recently, a new look for Bayesian-based pansharpening has been introduced in [32], which reported that CS and MRA-based methods could be derived from the Bayesian method by adjusting weights for details injection and spectral preservation. Khademi *et al.* [73] incorporated an adaptive image prior into a Bayesian model. Guo *et al.* [33] integrated the deep convolutional networks constrains and anisotropic total variational prior into a Bayesian model for pansharpening. From a sparse signal reconstruction point of view, Li and Yang [34] proposed a compressive sensing-based pansharpening. Ghahremani and Ghassemian [35] combined Ripplet transform and compressive sensing for pansharpening. Vicinanza *et al.* [36] explored a details self-similarity to alleviate the limitation of sparse representation to estimate the missing details. Tang *et al.* [37] presented a local-constrained sparse representation for pansharpening to keep data locality during the process of sparse representation. Yin *et al.* [38] injected details into a super-resolved MS image obtained by local adaptive sparse representation. To overcome the blurring results in the fused outcome obtained by conventional sparse representation, a convolutional sparse coding-based method combined with cartoon-texture decomposition was proposed in [39]. Fei *et al.* [40] incorporated convolutional sparse coding into details injection model to solve the limitations of traditional sparse representation-based methods, as the details preservation and the high sensitivity to misregistration.

B. DL-Based Pansharpening Methods

Inspired by DL-based image super-resolution [41], DL technologies have been strongly explored for pansharpening. Assuming the relationship between LRMS and HRMS images agrees with that between LR PAN and HR PAN images, Huang *et al.* [3] utilized stacked sparse denoising auto-encoder to learn the relationship between LR PAN and HR PAN images for pansharpening. Every band was independently reconstructed based on the learned relationship. Masi *et al.* [42] following the idea of the super-resolution with convolutional neural network (SRCNN) [41] proposed the first convolutional neural network (CNN) for pansharpening. Wei and Yuan [43] integrated deep residual learning (RL) into CNN. Yang *et al.* [44] presented PanNet, which processed LRMS and PAN images in a high-pass filtering domain to retain spatial information. Zhang *et al.* [45] proposed a bi-direction pyramid network for pansharpening. Yuan *et al.* [46] utilized multiscale features and RL for pansharpening. Scarpa *et al.* [47] explored a target-adaptive network. He *et al.* [48] exploited the detail injection model in the CNN architecture. Adopting two streams to separately extract information from LRMS and PAN images, Liu *et al.* [49] fused the information in the feature domain to reconstruct the HRMS image. Supposing the bands in the MS image were independent, Liu *et al.* [2] proposed a band-independent encoder–decoder network (BIEDN) to independently reconstruct each MS band. Wu *et al.* [50] utilized a proximal deep injection term to measure the distance between the output of a deep model and the desired HRMS image, which is regarded as a regularization term for the variational model. Fu *et al.* [51] explored grouped dilated convolution to extract multiscale details for pansharpening. Wu *et al.* [52] combined variational optimization and DL to take advantage of both technologies. Deng *et al.* [53] combined CS as well as MRA with deep neural networks proposing a detail injection-based deep CNN for pansharpening.

Although [3] and [2] can process MS images with different numbers of bands in a single model by independently reconstructing each band, the spectral discrepancy and dependency were neglected in both the methods, which led to spectral distortions in the reconstructed HRMS images. Different

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 12:30:08 UTC from IEEE Xplore. Restrictions apply.

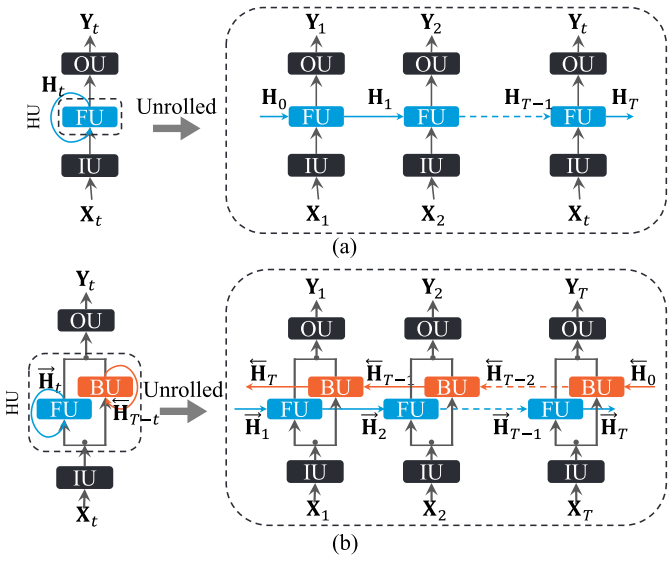


Fig. 2. Illustration of (a) RNN and (b) BiRNN. IU, HU, and OU refer to the input, the hidden, and the output unit (OU), respectively. FU and BU denote the forward and the backward unit, respectively.

from [2], [3], ArbRPN processes MS images as a union by considering the spectral dependency among bands. Benefiting from the capability of capturing the spectral dependency, our ArbRPN can achieve better spectral preservation in the reconstruction of HRMS images.

III. Methodology

A. Network Architecture

The proposed ArbRPN network has been designed following three properties: 1) the dominant spectral information in each band of the reconstructed HRMS should derive from the LRMS image, while the dominant high-frequency spatial details should derive from the PAN image; 2) the spectral discrepancy and dependency among the MS bands should be captured and utilized by the network, thus retaining the spectral features of the LRMS image into the reconstructed HRMS image; and 3) the high-frequency spatial information from the PAN image should be injected into each reconstructed band according to the spectral dependency between each band and the PAN image.

Following these three properties, we adopt first the global and local RL to ensure that the dominant spectral information of the reconstructed band originates from the corresponding band of the LRMS image. The high-frequency spatial information from the PAN image is injected into each band by fusing the features extracted from the PAN image into the hidden states. Afterward, to capture the spectral dependency of the LRMS bands, a BiRNN-based architecture is utilized to embed this latter into each hidden state. It is worth to be remarked that the RNN [4] can also be used to embed the spectral dependency, but, in that case, the hidden state only collects information before the current time step [as shown in Fig. 2(a)], which means that each hidden state only captured the spectral dependency between the current and the previous bands. Therefore, the RNN cannot make full use of the spectral dependency for MS images. Different from RNNs that only have FUs, BiRNNs [54] have both FUs and BUs in the hidden unit (HU), as shown in Fig. 2(b). Thus, the spectral information not only from previous but also from subsequent bands can be collected by the forward recurrences [i.e., the blue arrows in Fig. 2(b)] and the backward recurrences [i.e., the orange arrows in Fig. 2(b)]. Consequently, a BiRNN-based architecture is utilized in our ArbRPN, which can fully explore the spectral dependency to obtain better performance. However, rather than using the vanilla BiRNN shown in Fig. 2(b), our ArbRPN introduces a sequential BiRNN illustrated in Fig. 3(a). Different from the parallel architecture of FU and BU in the vanilla BiRNN, BU is stacked over the FU in our ArbRPN. That is because the sequential architecture of FU and BU in our sequential BiRNN leads to a deeper network than the vanilla BiRNN while keeping the same amount of parameters, which facilitates the performance, see Section IV-C. Besides, the sequential BiRNN architecture of our ArbRPN can be viewed as a Seq2Seq [55] model (see Appendix A for further details). Specifically, the processing under the backward recurrences can be regarded as the encoder in the Seq2Seq model, which is used to extract band features and to embed the spectral dependency among different bands. The processing over the forward recurrences can be viewed as a decoder in the Seq2Seq model, which is used to reconstruct the bands of the HRMS image. Finally, to account for the last property, we explicitly inject PAN features, h_{pan}, into each forward and backward recurrences for the reconstruction of each band of the HRMS image. The PAN features will be fused into the hidden states by both FU and BU, and, then, the injected details will be adaptively selected by the OU according to the captured spectral dependence, which is implicitly embedded into the hidden states through forward and backward recurrences. If we add the PAN features only in the initial hidden state for details injection, the PAN features will be forgotten in the latter time step, especially for MS images with a large number of bands, because our ArbRPN has no long-term memory.

The reasons why not using long-term memory are as follows.

1) As shown in Fig. 1, although a spectral overlap between adjacent bands of MS images exists, it is usually negligible between bands located at very different spectral wavelengths. Therefore, the performance gain in capturing the dependence of long distant bands is likely to be limited. Indeed, in our pre-experiments, we found that convolutional long short time memory (ConvLSTM) [56] and convolutional gate recurrent unit (ConvGRU) [57] only achieve slightly better results than the RNN while having much more parameters and a higher computation burden than the RNN.
2) The main goal of this article is to pansharpen MS images with any number of bands by using a single model finding a solution to train the network using MS samples with different numbers of bands, rather than exploring the best sequential model for pansharpening.

B. Solution

For convenience, we summarize the notation before elaborating on the proposed method. $\mathbf{X} \in \mathcal{R}^{h \times w \times B}$ is the LRMS

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 12:30:08 UTC from IEEE Xplore. Restrictions apply.

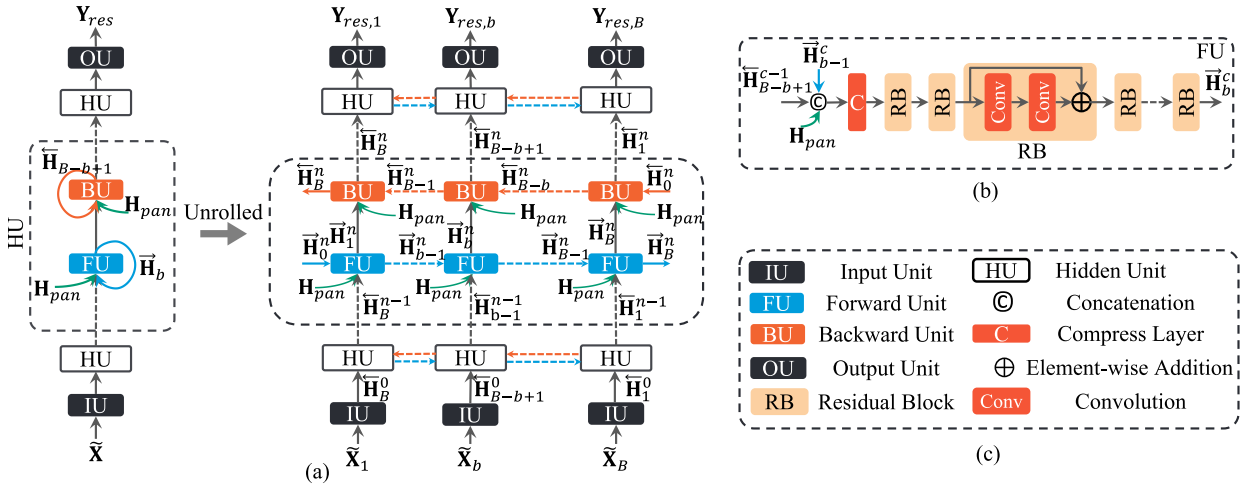


Fig. 3. Illustration of (a) proposed ArbRPN, (b) forward unit (FU), while the structure of the backward unit (BU) is the same as FU but with different inputs, and (c) short description of the modules in (a) and (b).

image with B bands. $\mathbf{X}_b$ refers to the bth band of $\mathbf{X}$. h and w are the height and width of an image, respectively. $\widetilde{\mathbf{X}}$ denotes the interpolated LRMS image. $\mathbf{Y} \in \mathcal{R}^{rh\times rw\times B}$ is the HRMS image. r is the scale ratio between HRMS/PAN and LRMS images. $\mathbf{X}_{\text{pan}} \in \mathcal{R}^{rh\times rw}$ is the PAN image. $\overrightarrow{\mathbf{H}}_b^n$ ($\overleftarrow{\mathbf{H}}_b^n$) denotes the hidden state at the bth forward (backward) recurrence in the nth HU. N denotes the total number of HUs.

As shown in Fig. 3, there are three modules in ArbRPN, i.e., the input unit (IU), the HU, and the OU. To improve the capacity, N HUs are stacked in the proposed approach. Given an LRMS image with B bands as input, the proposed ArbRPN will be unrolled to B time steps according to the number of bands in the LRMS image. Then, in each recurrence, a band of the HRMS image will be reconstructed by global RL. Thus, our ArbRPN is able to predict the residual between the GT and the interpolated LRMS image. Hence, the fusion equation of the proposed approach is as follows:

$$\begin{aligned}\widehat{\mathbf{Y}}_b &= \widetilde{\mathbf{X}}_b + \mathbf{Y}_{\text{res},b}, \quad b = B, \ldots, 2, 1\\ \mathbf{Y}_{\text{res},b} &= f_{\text{rpn},b}\left(\widetilde{\mathbf{X}}, \mathbf{X}_{\text{pan}}\right)\end{aligned} \tag{2}$$

where b is the band index, $\widetilde{\mathbf{X}}_b$ is the bth band of the interpolated LRMS image, including the low-frequency component in the reconstructed HRMS image, $\mathbf{Y}_{\text{res},b}$ is the residual band outputted by the ArbRPN containing the injected details in the reconstructed HRMS image, $\widehat{\mathbf{Y}}_b$ is the bth band of the reconstructed HRMS image, and $f_{\text{rpn},b}(\cdot)$ denotes a function, depending on the band index, getting the residuals in the proposed ArbRPN.

To utilize the spectral dependency rather than processing each band independently, we embed first the spectral dependency among the current band $\mathbf{X}_b$ and the previous bands $\mathbf{X}_s (s < b)$ into the forward hidden states, $\overrightarrow{\mathbf{H}}_b^n$, by the forward recurrence in the HU, which can be formulated as follows:

$$\overrightarrow{\mathbf{H}}_b^n = f_{fu}\left(\overrightarrow{\mathbf{H}}_{b-1}^n, \overleftarrow{\mathbf{H}}_{B-b+1}^{n-1}, \mathbf{H}_{\text{pan}}\right) \tag{3}$$

where $f_{fu}(\cdot)$ denotes the function of the FU, $\mathbf{H}_{\text{pan}}$ represents the features extracted from the PAN image, $\mathbf{X}_{\text{pan}}$, by a convolutional layer, and $\overleftarrow{\mathbf{H}}_{B-b+1}^{n-1}$ is the backward hidden state at the $(B-b)$th backward recurrence in the $(n-1)$th HU. The backward hidden states fed into the first HU are obtained by the IU, which is formulated as follows:

$$\overleftarrow{\mathbf{H}}_{B-b+1}^0 = f_{iu}\left(\widetilde{\mathbf{X}}_b\right), \quad b = 1, 2, \ldots, B \tag{4}$$

where $f_{iu}(\cdot)$ denotes the function of the IU, which is constituted by a single convolutional layer. The initial hidden states for both forward and BUs in all the HUs are assigned to the PAN features, i.e.,

$$\overrightarrow{\mathbf{H}}_0^n = \mathbf{H}_{\text{pan}}, \quad \overleftarrow{\mathbf{H}}_0^n = \mathbf{H}_{\text{pan}}, \quad n = 1, 2, \ldots, N. \tag{5}$$

Afterward, a backward recurrence is used to embed the spectral dependency between the current band $\overrightarrow{\mathbf{H}}_b^n$ and the subsequent bands $\overrightarrow{\mathbf{H}}_s^n (s > b)$, which can be formulated as follows:

$$\overleftarrow{\mathbf{H}}_{B-b+1}^n = f_{bu}\left(\overrightarrow{\mathbf{H}}_b^n, \overleftarrow{\mathbf{H}}_{B-b}^n, \mathbf{H}_{\text{pan}}\right) \tag{6}$$

where $f_{bu}(\cdot)$ denotes the function of the BU. Through a cycle of all the forward and backward recurrences, the spectral dependency among all the bands is embedded into each backward hidden state, $\overleftarrow{\mathbf{H}}_b^n (n > 1)$. Finally, the OU, built by a single convolutional layer, is utilized to reconstruct the residual bands from the last HU

$$\mathbf{Y}_{\text{res},b} = f_{ou}\left(\overleftarrow{\mathbf{H}}_{B-b+1}^N\right) \tag{7}$$

where $f_{ou}(\cdot)$ denotes the function of the OU.

C. Hidden Unit

As shown in the unrolled architecture of the HU in Fig. 4, each HU obtains the output $\{\overleftarrow{\mathbf{H}}_B^n, \overleftarrow{\mathbf{H}}_{B-1}^n, \ldots, \overleftarrow{\mathbf{H}}_1^n\}$ by taking as input the previous HU's output, $\{\overleftarrow{\mathbf{H}}_B^{n-1}, \overleftarrow{\mathbf{H}}_{B-1}^{n-1}, \ldots, \overleftarrow{\mathbf{H}}_1^{n-1}\}$. Specifically, the processing routes for different spectral bands are also shown in Fig. 4 by exploiting different colors. In the HU, forward hidden states of all the bands are obtained first by the forward recurrence formulated by (3). Then, the output of each band is obtained by the backward recurrence formulated

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 12:30:08 UTC from IEEE Xplore. Restrictions apply.

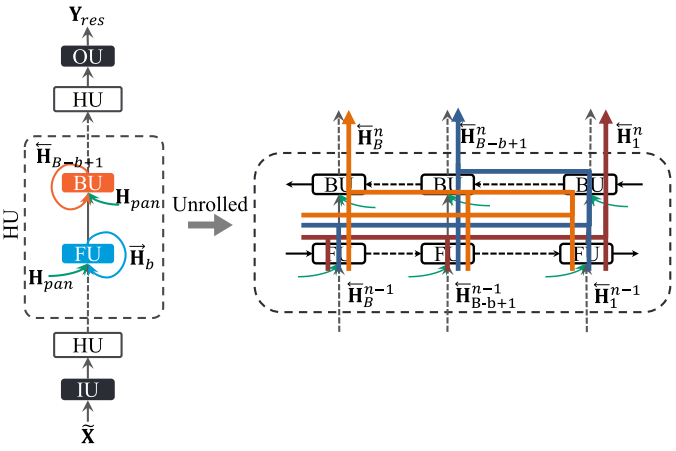


Fig. 4. Illustration of the processing routes for different spectral bands in the HU. To clearly show the processing routes, we removed some components in the unrolled network.

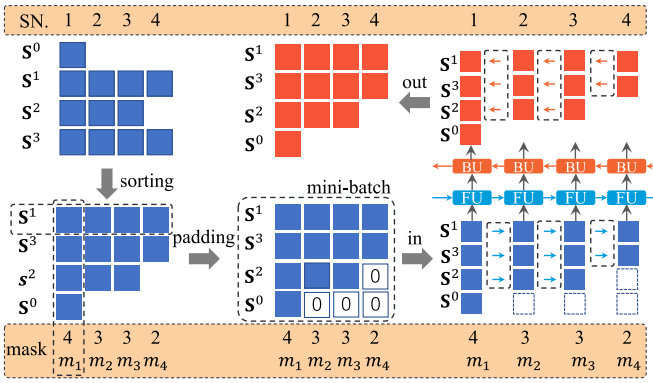


Fig. 5. Schematic of the proposed mask-training. For simplicity, only a cycle of the forward and backward recurrence in the HU is depicted in this figure. (Top row) The band serial number (SN). (Bottom row) The mask for each band. The process of mask-training is illustrated in the middle area.

by (6). Therefore, for the specific processing of the bth band in the nth HU, the output can be obtained by combining (6) and (3). Thus, we have

$$\begin{aligned}\overleftarrow{\mathbf{H}}^{n}_{B-b+1} &= f_{bu}\left(\overrightarrow{\mathbf{H}}^{n}_{b}, \overleftarrow{\mathbf{H}}^{n}_{B-b}, \mathbf{H}_{\text{pan}}\right) \\ &= f_{bu}\left(f_{fu}\left(\overrightarrow{\mathbf{H}}^{n}_{b-1}, \overleftarrow{\mathbf{H}}^{n-1}_{B-b+1}, \mathbf{H}_{\text{pan}}\right), \overleftarrow{\mathbf{H}}^{n}_{B-b}, \mathbf{H}_{\text{pan}}\right)\end{aligned} \tag{8}$$

where $\overleftarrow{\mathbf{H}}^{n-1}_{B-b+1}$ is the output of the bth band from the previous HU, $\overrightarrow{\mathbf{H}}^{n}_{b-1}$ is the forward hidden state from the previous forward time step obtained by (3), $\overleftarrow{\mathbf{H}}^{n}_{B-b}$ is the backward hidden state from the previous backward time step obtained by (6), $\overrightarrow{\mathbf{H}}^{n}_{0}$ and $\overleftarrow{\mathbf{H}}^{n}_{0}$ are the initial hidden states for both forward and backward, respectively, provided by (5), and $\mathbf{H}_{\text{pan}}$ are the PAN features extracted by a convolutional layer.

D. Forward/BUs

The architecture of the FU is illustrated in Fig. 3(b), which consists of a compressive layer and L residual blocks (RBs) [58], [59]. The architecture of the BU is the same as FU. The compressive layer, implemented by a 1×1 convolutional layer, is utilized to fuse the PAN features, the forward hidden state, and the backward hidden state. RBs are used to achieve the local RL and to refine the fused features. Therefore, the equation of the FU is as follows:

$$\mathbf{H}_{f} = f_{c}\left(\left[\overrightarrow{\mathbf{H}}^{n}_{b-1}, \overleftarrow{\mathbf{H}}^{c-1}_{B-b+1}, \mathbf{H}_{\text{pan}}\right]\right) \tag{9}$$

$$\overrightarrow{\mathbf{H}}^{n}_{b} = f_{rb,L}\left(\cdots, f_{rb,2}\left(f_{rb,1}\left(\mathbf{H}_{f}\right)\right)\cdots\right) \tag{10}$$

where $[\cdot]$ denotes the concatenation along the channel direction, $f_{c}(\cdot)$ represents the function of the compressive layer, $\mathbf{H}_{f}$ indicates the fused features, and $f_{rb,l}(\cdot)$ is the lth RB in the FU.

E. Mask-Training

In existing DL-based pansharpening methods, MS images with different numbers of bands cannot be grouped into a mini-batch for training. Inspired by mask language models [60] in natural language processing, we introduce a mask-based training method called mask-training to solve this problem. Specifically, the illustration of mask-training is shown in Fig. 5. First of all, M (equal to 4 in Fig. 5) samples of LRMS, HRMS, and PAN images are collected from the training set. $\mathbf{S}^{i}$ denotes the ith sample in the set $\mathbb{S}$ of the collected training samples. Afterward, these samples are sorted in descending order according to the number of bands in each LRMS image.[1] The maximum number of bands is denoted as B. Then, for the bth ($1 \leq b \leq B$) band, a mask is used to record how many MS images have the specific bth band. The value of the mask for the bth band (indicated as m_b) is obtained by

$$m_{b} = \text{Count}(\mathbb{S}, b) \tag{11}$$

where $\text{Count}(\mathbb{S}, b)$ is the function that counts how many LRMS images in the set $\mathbb{S}$ have the bth band. For example, three LRMS images have the third band in the collected images in Fig. 5, thus, the mask number is three for the third band, i.e., $m_3 = 3$. Since only MS images with a constant number of bands can be grouped in a mini-batch, zeros are padded in LRMS images to make their numbers of bands equal to the maximal number of bands, thereby enabling these LRMS images to be grouped into a mini-batch. After the mini-batch is fed into our ArbRPN, to save the computational burden and to avoid the effects of padded zeros [61], we remove the padded zeros based on the mask. At the bth recurrence, only the hidden states of the previous m_{b+1} samples (i.e., the hidden states of $\{\mathbf{S}^{1}, \mathbf{S}^{2}, \ldots, \mathbf{S}^{m_{b+1}}\}$) are transferred to the next recurrence. After the processing of all the bands, the reconstructed bands are recombined to produce the reconstructed HRMS image. At last, the loss function is applied to the reconstructed HRMS image to supervise the training of the proposed approach.

Like [2], the l_1 norm is adopted as loss function in our approach. Since HRMS images are padded with zeros to construct a mini-batch, while the padded zeros are removed

[1]The HRMS and PAN images are also sorted. For simplicity, only the sorting of LRMS images is shown in Fig. 5.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 12:30:08 UTC from IEEE Xplore. Restrictions apply.

in the LRMS images, the padded zeros in the HRMS images have to be removed to calculate the loss, too. Specifically, the loss function is defined as follows:

$$\text{loss} = \frac{1}{M}\sum_{i=1}^{M}\frac{\left\|\hat{\mathbf{Y}}^i - \mathbf{Y}^i\right\|_1}{B_i H W} \tag{12}$$

where $\hat{\mathbf{Y}}^i$ and $\mathbf{Y}^i$ are the ith reconstructed HRMS image and GT, respectively, M is the number of the collected samples that are used to construct the mini-batch, H and W are the height and width of the HRMS image, B_i represents the number of bands in the ith HRMS image, which is used to get the average loss along all the bands in the reconstructed HRMS images. Indeed, since the HRMS images with more bands lead to a bigger total loss, the average loss along all the bands seems a more suitable choice.

IV. Experiments

A. Datasets, Training, and Architecture Details

1) Dataset Details: Data acquired by the QB and WorldView-2 (WV2) sensors are used in our experiments. MS and PAN images are cropped first to 256×256 and 1024×1024, respectively. Then, for each sensor, these cropped images are divided into training set and test set with a ratio of about 9:1. Specifically, 403 and 514 image pairs are used as training samples for QB and WV2, respectively. Forty-four samples are used as the test set for QB and 57 samples are used as test set for WV2. The results reported in all the tables of this article are the average results on the corresponding test sets. In each training set, ten image pairs are randomly chosen as the validation set. During the training stage, we also use random flipping and rotating augmentation to get 393×2 and 504×2 samples for QB and WV2, respectively.

Akin to [63], Wald's protocol [64] with modulation transfer function (MTF)-based spatial degradation is utilized to generate training samples. The MTF functions adopted in this article are the same as those in [63]. For more details about MTF-based degradation, please refer to [15]. The corresponding degradation is also applied to the test set to generate test samples at reduced resolution (RR), while the original test set is adopted as test samples at full resolution (FR). Two types of datasets are synthesized for the training. The first type (marked as FIX-N) only consists of MS images with the same number, i.e., N, of bands, which is used to train the models for LRMS images with a fixed number of bands. The second type (marked as MIX-N) consists of samples randomly selected from FIX-1 to FIX-N with an equal proportion, which is used to train a model for LRMS images with any number of bands. For performance testing on MS images with the fixed number of bands, the original test sets at both RR and FR are adopted. For performance testing on MS images with different numbers of bands, four test sets (i.e., from FIX-1 to FIX-4) from QB and eight test sets (i.e., from FIX-1 to FIX-8) from WV2 are employed.

Specifically, FIX-N is generated by randomly selecting N bands from the original MS images. Since the maximum number of bands in MS images captured by QB is four, $N \leq 4$ with $N \in \mathcal{Z}^+$ is set for FIX-N from QB. Similarly, $N \leq 8$ with $N \in \mathcal{Z}^+$ is set for FIX-N from WV2. Regarding MIX-N, it is generated by randomly selecting samples from FIX-1 to FIX-N with an equal proportion, i.e.,

$$\Omega_{\text{MIX-N}} = S\left(\Omega_{\text{FIX}-1}, \frac{1}{N}\right) \cup S\left(\Omega_{\text{FIX}-2}, \frac{1}{N}\right) \cup \cdots \cup S\left(\Omega_{\text{FIX-N}}, \frac{1}{N}\right) \tag{13}$$

where $\Omega_{\text{FIX-N}}$ refers to the FIX-N dataset, $\Omega_{\text{MIX-N}}$ refers to the MIX-N dataset, $S[\Omega_{\text{FIX-N}}, (1/N)]$ refers to the action of taking $1/N$ samples from the $\Omega_{\text{FIX-N}}$ set in a random way, and $\cup$ denotes the operation of the union.

Finally, the radiometric resolution of QB and WV2 data is 11 bit. Thus, we normalize the values of pixels in MS and PAN images to [0,1] dividing them by 2047.

2) Training Details: In the training stage, we exploit Adam to optimize the proposed network using 10^6 iterations with an initial learning rate of 0.0001. The learning rate halves every 2×10^5 iterations. The batch size and the patch size of LRMS images are set to 8 and 16 for training, respectively.

3) Architecture Details: The number of HUs and the number of RBs in a HU are two key hyper-parameters controlling the capacity of our ArbRPN. We set the number of HUs and the number of RBs in the FU/BU to 5 and 3, i.e., $N = 5$ and $L = 3$, respectively, to achieve similar parameters compared to [49]. Noteworthy, our ArbRPN can achieve better performance by using higher N and L, as shown in Section IV-C. Except for the convolution layers in the OU, the number of output channels for all the convolutional layers is set to 64. The number of the output channels for the convolutional layer in the OU is set to 1. The kernel size, stride, and padding of all the convolutional layers except for the compressive layer in the FU and the BU are set to 3×3, 1, and 1, respectively. The kernel size, stride, and padding of the compressive layer are set to 1, 1, and 0, respectively. The proposed method is implemented in Python3.7 based on Pytorch 1.3. We use in this article standard implementations of traditional methods that can be freely downloaded [63], [65]. Other DL-based comparative methods are reimplemented according to the corresponding papers. Networks in this article are trained on a desktop with a GTX-1080Ti GPU and tested on a desktop with a GTX-1060 GPU.

B. Comparison With SOTA Methods

To validate the performance of the proposed method, we compare it with several SOTA methods, including traditional- and DL-based methods at both RR and FR. Four reference-based metrics, including the correlation coefficient (CC) [66], the erreur relative globale adimensionnelle de synthèse (ERGAS) [67], the spectral angle mapper (SAM) [68], and the Q2n index (i.e., Q4 for four bands and Q8 for eight bands) [69], are used to assess all the methods at RR. Ideal values are 1 for CC and Q2n, and 0 for ERGAS and SAM. Instead, at FR, a nonreference-based metric, i.e., the quality without reference (QNR) [70], is utilized for comparison purposes consisting of the combination of the spectral distortion

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 12:30:08 UTC from IEEE Xplore. Restrictions apply.

TABLE I

RR AND FR RESULTS OF MS IMAGES WITH A FIXED NUMBER OF BANDS (I.E., FIX-4) ACQUIRED BY THE QB AND WV2 SENSORS. THE BEST RESULTS ARE IN BOLDFACE, WHILE THE SECOND-BEST RESULTS ARE UNDERLINED

Methods	QB							WV2						
	RR Test				FR Test			RR Test				FR Test		
	CC↑	Q4↑	ERGAS↓	SAM↓	$D_\lambda\downarrow$	$D_s\downarrow$	QNR↑	CC↑	Q8↑	ERGAS↓	SAM↓	$D_\lambda\downarrow$	$D_s\downarrow$	QNR↑
EXP	0.7639	0.5599	3.0160	3.4764	**0.0000**	0.1063	0.8937	0.7925	0.6080	7.4350	7.7174	**0.0000**	0.0593	0.9407
PCA	0.8100	0.6749	2.6586	3.3963	0.0304	0.1271	0.8474	0.7649	0.6674	7.2024	9.5091	0.0582	0.1511	0.8004
IHS	0.7899	0.6357	2.6893	3.6241	0.0975	0.1692	0.7523	0.8777	0.7338	6.1000	7.9746	0.0842	0.1286	0.7996
BDSD	0.8783	0.7945	1.9434	2.5357	0.0203	0.0429	0.9379	0.8898	0.8156	5.2645	8.0761	0.0821	0.1049	0.8221
MTF-GLP	0.8802	0.7738	2.0508	2.7097	0.0695	0.1241	0.8162	0.9024	0.8179	5.1425	7.3228	0.0884	0.1501	0.7758
C-GSA	0.8848	0.7895	1.9301	2.5559	0.0668	0.1513	0.7944	0.9069	0.8230	5.1099	7.5766	0.0639	0.0944	0.8488
BDSD-PC	0.8799	0.7946	1.9321	2.5204	0.0232	0.0726	0.9064	0.9018	0.8383	4.8512	7.3370	0.0466	0.0414	0.9156
MTF-GLP-FS	0.8798	0.7895	1.9480	2.5129	0.0395	0.0845	0.8799	0.9020	0.8222	5.0740	7.5252	0.0782	0.0823	0.8472
MTF-GLP-HPM-H	0.8875	0.7973	1.9499	2.7653	0.0970	0.1304	0.7863	0.9107	0.8249	4.8906	6.9665	0.1037	0.0973	0.8112
PNN	0.9161	0.8478	1.5730	1.9869	0.0342	0.0769	0.8949	0.9525	0.9071	3.4059	5.3662	0.0525	0.0387	0.9127
DRPNN	0.9427	0.8902	1.2508	1.6302	0.0254	0.0478	0.9287	0.9689	0.9316	2.7088	4.3973	0.0530	0.0478	0.9038
PANNET	0.9126	0.8467	1.5911	2.0125	0.0244	0.0678	0.9113	0.9537	0.9088	3.3808	5.4102	0.0604	0.0497	0.8939
DICNN1	0.9172	0.8486	1.5703	1.9790	0.0282	0.0787	0.8982	0.9516	0.9060	3.4427	5.4194	0.0518	0.0445	0.9079
MSDCNN	0.9219	0.8554	1.5066	1.8849	0.0291	0.0725	0.9034	0.9575	0.9148	3.2106	5.0901	0.0525	0.0403	0.9112
TFNET	0.9502	0.9015	0.1620	0.5297	0.0210	0.0419	0.9398	0.9690	0.9320	2.7024	4.4256	0.0480	0.0379	0.9176
ArbRPN* (ours)	**0.9662**	**0.9291**	**0.9510**	**0.3240**	0.0018	0.0406	**0.9432**	**0.9754**	**0.9408**	**0.4039**	**0.0059**	0.0317	**0.0261**	**0.9442**
BIEDN	0.9284	0.8666	1.4303	1.8386	0.0325	**0.0403**	0.9285	0.9621	0.9211	3.0238	4.8966	0.0823	0.0467	0.8761
ArbRPN (ours)	0.9324	0.8745	1.2731	1.6443	0.0227	0.0438	0.9353	0.9739	0.9386	0.4871	0.1316	0.0410	0.0267	0.9344

index, D_λ, and the spatial distortion index, D_s. Optimal values are 0 for both the distortion indexes and 1 for the QNR.

The SOTA methods used for comparison are listed below.

1) Traditional methods.
   a) EXP: Interpolation-based method applied to the MS image based on a polynomial kernel with 23 coefficients [62].
   b) PCA: Principle component analysis for pansharpening [71].
   c) IHS: IHS transformation-based method for pansharpening [5].
   d) BDSD: BDSD injection model for pansharpening [72].
   e) MTF-GLP: GLP [62] method with low-pass filter matched with the MTF of the MS sensor [15].
   f) C-GSA: Context-adaptive estimation of injection coefficients applied to the GSA approach for pansharpening [21].
   g) BDSD-PC: BDSD method, including physical constraints [12].
   h) MTF-GLP-FS: MTF-GLP method with a full-scale regression-based injection coefficient estimation [22].
   i) MTF-GLP-HPM-H: MTF-GLP with high-pass modulation [19] and haze correction [24].
2) DL-based methods.
   a) PNN: Pansharpening by CNN [42].
   b) DRPNN: Convolutional network with deep RL for pansharpening [43].
   c) PANNET: Global RL with high-pass domain trained network for pansharpening [44].
   d) DICNN1: Detail injection-based CNN for pansharpening [48].
   e) MSDCNN: Multiscale and multiple depth network for pansharpening [46].
   f) TFNET: Two-stream fusion network for pansharpening [49].
   g) BIEDN: BIEDN for pansharpening [2].

1) Results on MS Images With a Fixed Number of Bands: Since most of the methods are proposed for MS images with a fixed number of bands, especially for DL-based approaches, we compare first our ArbRPN with SOTA methods on MS images with a fixed number of bands. Two models of ArbRPN are trained. One marked as ArbRPN*, which is trained using MS images with a fixed number of bands, i.e., FIX-N, while the other, marked as ArbRPN, is trained using MS images with different numbers of bands, i.e., MIX-N. The quantitative results of these comparative methods are shown in Table I. It is worth remarking that our ArbRPN* achieves the best results in all the metrics except D_λ and D_s on both QB and WV2 datasets, thus demonstrating the potential of the proposed method for pansharpening. The EXP achieves the ideal result in D_λ because the result of EXP is regarded as the reference to measure the spectral distortion as done in [63]. Although EXP and BIEDN, respectively, achieve better results than ArbRPN* on QB regarding D_λ and D_s, the QNR of ArbRPN* is the highest, which demonstrated ArbRPN* can achieve a better tradeoff between spectral and spatial distortion than these methods. Our ArbRPN, i.e., trained on MIX-N, also outperforms other methods, but only on WV2 data. It is because MIX-8 of WV2 has sufficient samples to learn the spectral dependency. For MIX-4 of QB, according to (13), the 25% of samples have one band, thus our ArbRPN cannot fully learn the spectral dependency due to the lack of sufficient training samples. More analysis about the generalization of our ArbRPN on MS images on different numbers of bands is presented in Section IV-D. Noteworthy, the QNR of the EXP is higher than all that of other methods except ArbRPN for the WV2 test case. This is because some samples in the test set have few details at FR. Thus, the EXP can leverage on an optimal spectral preservation, testified by a D_λ index equal to 0, as well as a reduced spatial distortion, see Table I, thereby leading to a high QNR.

To show the visual effects of the results obtained by our ArbRPN, we compare the outcomes obtained by the benchmark at both reduced and FR. For the sake of brevity, only

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 12:30:08 UTC from IEEE Xplore. Restrictions apply.

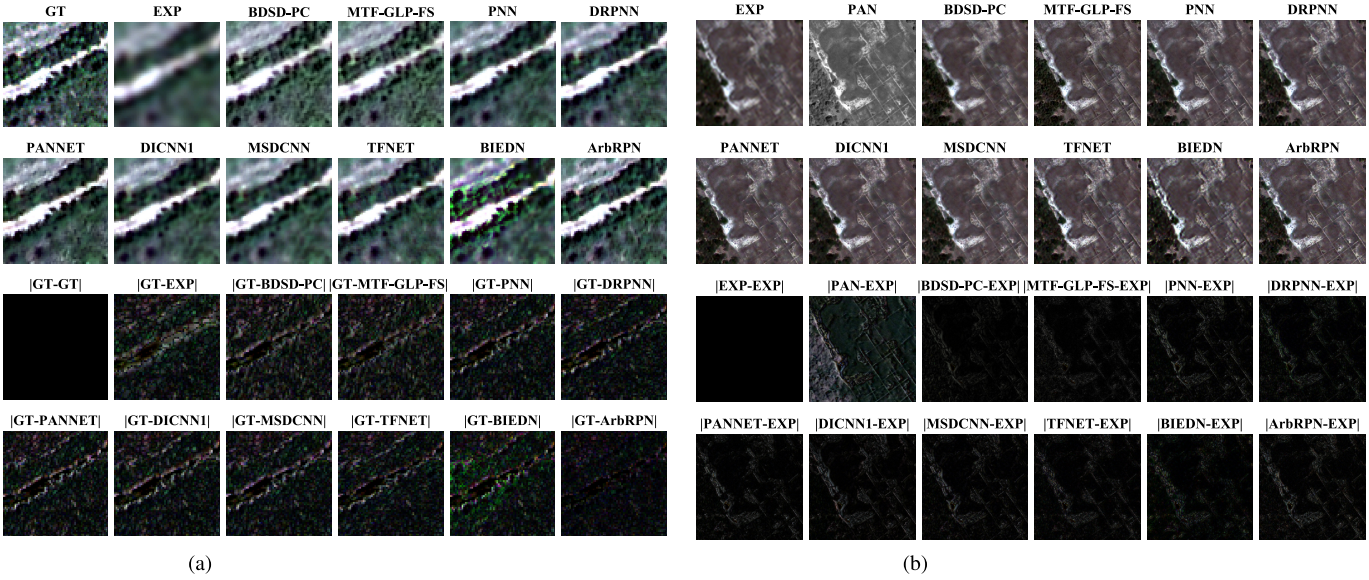


Fig. 6. Comparison of results obtained by models for MS images with a fixed number of bands acquired by the QB sensor. The two bottom rows in (a) are residual images between reconstructed images and the GT. The two bottom rows in (b) are residual images between reconstructed images and the EXP [62]. (a) Results at RR. (b) Results at FR.

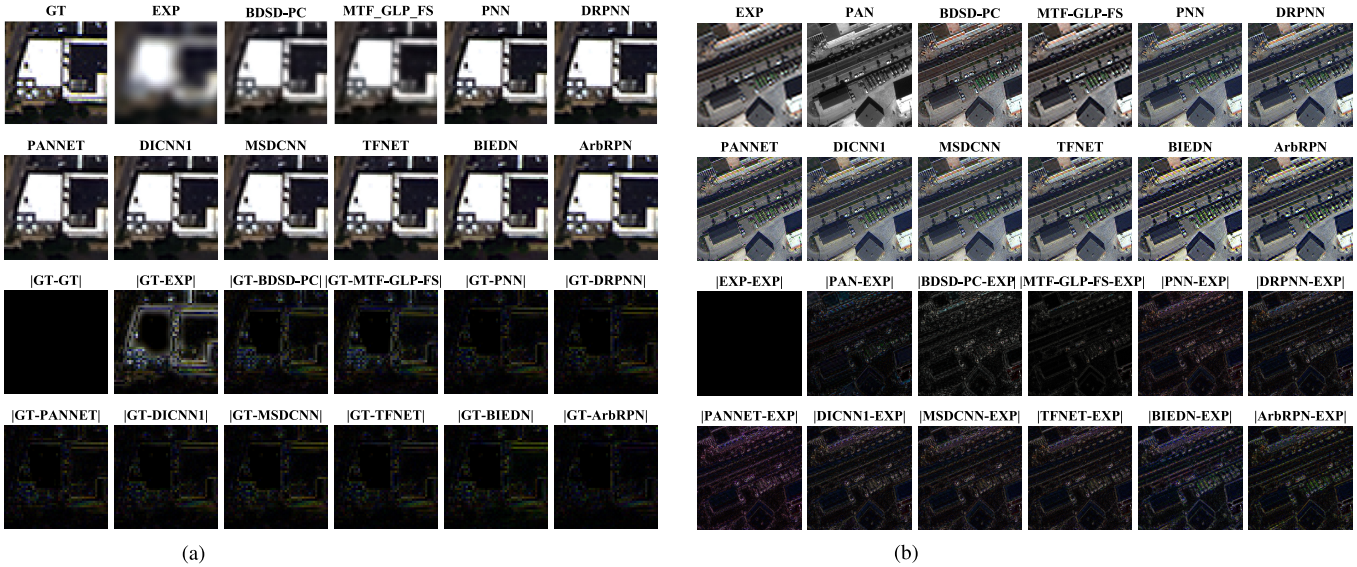


Fig. 7. Comparison of results obtained by models for MS images with a fixed number of bands acquired by the WV2 sensor. (a) Results at RR. (b) Results at FR.

a subset of the methods (always including all the DL-based ones) reported in the tables is depicted for visual comparison. This is done to reduce the number of figures in the article. As shown in Fig. 6(a), for QB results at RR, it can be found that BDSD-PC and MTF-GLP-FS achieve blur results than that of the compared DL-based methods. Many colorful pixels in the BIEDN residual image instead indicate a relevant spectral distortion. By contrast, our ArbRPN restores clearer edges and achieves better spectral preservation. The residual image also shows that the reconstructed HRMS image by our ArbRPN is the closest one to the GT. As shown in Fig. 6(b), for QB results at FR, the residual images between the results obtained by the EXP and the other methods are shown to aid the visual inspection. Since the EXP does not inject details from the PAN image into the LRMS image, the result of the EXP shows similar spectral characteristics with respect to the LRMS image, which can be used as a reference to evaluate spectral preservation. The residual images show that BDSD-PC and MTF-GLP-FS inject fewer details than the compared DL-based methods. Inside the DL-based class, we can find that the residual image of our method has more edges than the other approaches indicating that more details and edges are injected into the LRMS image. An interesting finding is that partial edges can be injected by these other methods and almost all the compared approaches hardly enhance textures in forest areas, see also Fig. 8(b). This phenomenon indicates that there is still room for improvement. As shown in Fig. 7(a), for WV2 results at RR, it can be found that EXP, BDSD-PC, and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 12:30:08 UTC from IEEE Xplore. Restrictions apply.

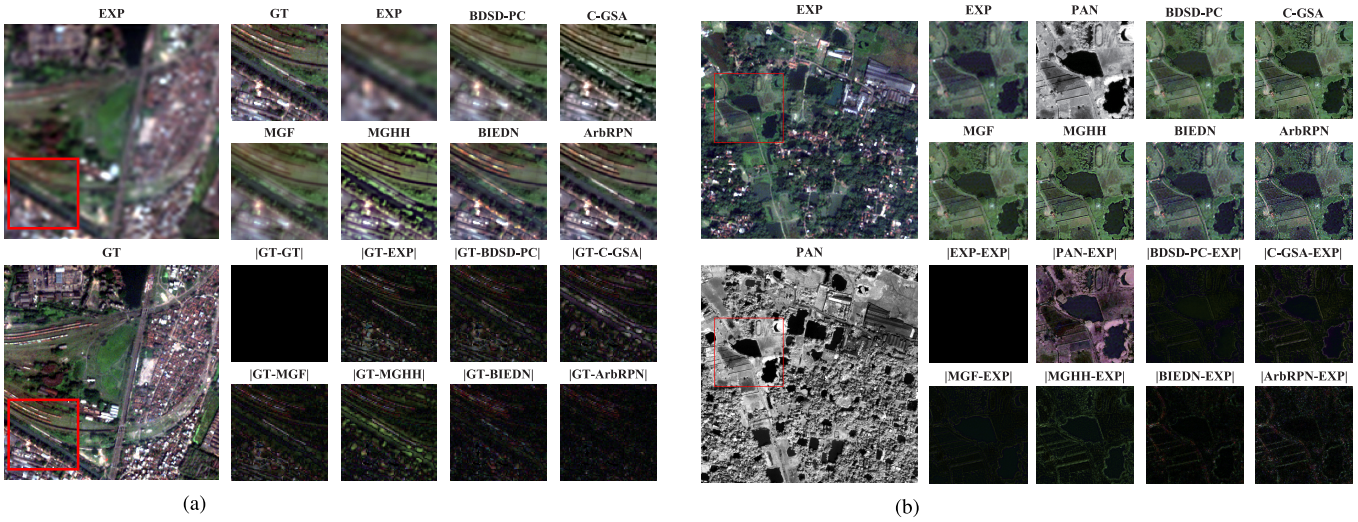


Fig. 8. Comparison of results obtained by models for MS images with various numbers of bands acquired by the QB sensor. MGF refers to MTF-GLP-FS. MGHH refers to MTF-GLP-HPM-H. (a) Results at RR. (b) Results at FR.

MTF-GLP-FS encounter more spectral distortions than the DL-based methods on the building area. The edges of the buildings are also not as clear as the results obtained by DL-based methods. For the DL-based methods, we can find that our ArbRPN gets lower errors than the other DL-based methods. The fewer colored pixels in the residual image in Fig. 7(a) also indicate that our ArbRPN has better spectral preservation than the other methods. As shown in Fig. 7(b), for WV2 results at FR, BDSD-PC and MTF-GLP-FS still inject fewer details than DL-based methods. The residual images obtained by PNN, PANNET, DICNN, MSDCNN, and BIEDN have a clear spectral distortion as the whole residual images present many light-red color pixels. Thanks to the consideration of the spectral discrepancy and dependency, our ArbRPN achieves better spectral preservation than the above-mentioned methods. Although the spectral preservation of our ArbRPN is slightly worse than the BDSD-PC, the MTF-GLP-FS, and the DRPNN, it injects more edges into the fused products than the other methods. Overall, our ArbRPN can achieve a better tradeoff between details enhancement and spectral preservation because of the ability to learn spectral discrepancy and dependency.

2) Results on MS Images With Different Numbers of Bands: To validate the effectiveness of the proposed network for MS images with arbitrary numbers of bands, we test our ArbRPN with test sets, including MS images with different numbers of bands. As for DL-based methods, since most of them cannot tackle MS images with arbitrary numbers of bands with a single model, only BIEDN [2] is used for comparison in this subsection. Besides, some traditional methods are also compared with our method. The quantitative results on MS with different numbers of bands are shown in Tables II and III. Regarding QB data, see Table II, our ArbRPN outperforms the other methods in almost all the metrics. Regarding WV2 data, see Table III, it is clear that our ArbRPN obtains the best results on all the tests except for FIX-1 and D_s and QNR on FIX-2. Indeed, on FIX-1, the BIEDN achieves a little better result than our ArbRPN. This is because BIEDN is trained by one-band MS images. Besides, BIEDN is trained with all the bands in FIX-8 using WV2 data, while our ArbRPN is trained with MIX-8. Therefore, according to (13), it can be found that the BIEDN has more training data than our ArbRPN. About the FR assessment, the EXP achieves the best performance while our ArbRPN achieves the second-best result for WV2 test cases. This is because the EXP is an interpolation method showing optimal spectral preservation according to the adopted metric. Anyway, this result is misleading. Indeed, even the spatial details have their spectral content and, thus, the simple interpolation cannot get the optimal spectral distortion. Moreover, the spatial distortion should be better considered in mixed scenarios to prevent the good ranking of interpolation methods that cannot enhance the spatial resolution. Indeed, it was already observed that when rural (instead of urban) scenarios are considered, we can easily get a paradox, i.e., having the best results obtained by the EXP when the QNR is used as a quality metric. Thus, the limits in the quantitative evaluation of the QNR index, already remarked in some relevant works [65], [69], are clear and researches are still ongoing to develop a reliable FR quality index.

To show the visual results tested on MS images with different numbers of bands, we show the natural color images (i.e., RGB bands) from four-band MS images in the QB test set and eight-band MS images in the WV2 test set. Fig. 8(a) shows the results at RR for the QB test case. Our ArbRPN has fewer errors and a reduced spectral distortion than the results obtained by the other methods demonstrating its superiority. The QB FR results are depicted in Fig. 8(b). Although BDSD-PC and MTF-GLP-FS have better performance in spectral preservation according to the fewer color pixels, they inject fewer details and edges than other methods. For MTF-GLP-HPM-H, it can find a clear spectral distortion (see the several green pixels in the residual image), which may result from an incorrect haze correction. Compared with the BIEDN, our ArbRPN injects more details and edges while

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 12:30:08 UTC from IEEE Xplore. Restrictions apply.

TABLE II

RESULTS OF THE COMPARED METHODS ON MS IMAGES WITH DIFFERENT NUMBERS OF BANDS FROM QB TEST SETS. '-' INDICATES THAT THERE IS NO RESULT CORRESPONDING TO THE METRIC DUE TO INVALID COMPUTATION. THE BEST RESULTS ARE IN BOLDFACE, WHILE THE SECOND-BEST RESULTS ARE UNDERLINED

Methods	RR Test				FR Test			RR Test				FR Test		
	CC↑	Q2n↑	ERGAS↓	SAM↓	$D_\lambda\downarrow$	$D_s\downarrow$	QNR↑	CC↑	Q2n↑	ERGAS↓	SAM↓	$D_\lambda\downarrow$	$D_s\downarrow$	QNR↑
	FIX-1							FIX-2						
EXP	0.7394	0.5292	3.1272	-	-	0.1061	-	0.7480	0.5351	3.2304	2.5764	**0.0000**	0.1038	0.8962
PCA	0.7023	0.5592	3.1400	-	-	0.2587	-	0.7602	0.6287	2.9692	2.6270	0.0503	0.1878	0.7723
IHS	0.7023	0.5592	3.1400	-	-	0.2587	-	0.7495	0.5940	2.9790	2.8010	0.0928	0.2115	0.7196
BDSD	0.8296	0.6494	2.7672	-	-	0.1336	-	0.8617	0.7673	2.2025	2.0299	0.0288	0.0724	0.9011
MTF-GLP	0.8744	0.7613	2.2198	-	-	0.1242	-	0.8728	0.7682	2.1471	1.9637	0.0846	0.1205	0.8070
C-GSA	0.6852	0.5697	6.3291	-	-	0.2953	-	0.8294	0.7198	2.4910	2.6256	0.1380	0.2072	0.6847
BDSD-PC	0.8346	0.6582	2.7291	-	-	0.0992	-	0.8673	0.7581	2.0398	1.8969	0.0269	0.0830	0.8924
MTF-GLP-FS	0.8836	0.7862	2.0406	-	-	0.0784	-	0.8829	0.7854	1.9199	1.8267	0.0484	0.0792	0.8766
MTF-GLP-HPM-H	0.2289	0.4955	597.75	-	-	0.1300	-	0.6176	0.6736	635.17	2.4307	0.1182	0.1156	0.7826
BIEDN	0.9298	0.8751	1.4922	-	-	0.0402	-	0.9294	0.8670	1.4684	1.3229	0.0266	**0.0442**	**0.9305**
ArbRPN (ours)	**0.9581**	**0.9180**	**1.1995**	-	-	**0.0397**	-	**0.9350**	**0.8780**	**1.2785**	**1.1475**	0.0263	0.0506	0.9252
	FIX-3							FIX-4						
EXP	0.7494	0.5422	3.0301	2.9368	**0.0000**	0.1131	0.8869	0.7639	0.5599	3.0160	3.4764	**0.0000**	0.1063	0.8937
PCA	0.7394	0.6152	2.8922	3.0372	0.0463	0.1789	0.7844	0.8100	0.6749	2.6586	3.3963	0.0304	0.1271	0.8474
IHS	0.7350	0.5960	2.8552	3.1163	0.0988	0.1876	0.7364	0.7899	0.6357	2.6893	3.6241	0.0975	0.1692	0.7523
BDSD	0.8656	0.7797	2.0249	2.2381	0.0187	0.0528	0.9298	0.8783	0.7945	1.9434	2.5357	0.0203	0.0429	**0.9379**
MTF-GLP	0.8684	0.7549	2.1576	2.4100	0.0810	0.1205	0.8102	0.8802	0.7738	2.0508	2.7097	0.0695	0.1241	0.8162
C-GSA	0.8685	0.7634	2.1001	2.3344	0.0962	0.1550	0.7661	0.8848	0.7895	1.9301	2.5559	0.0668	0.1513	0.7944
BDSD-PC	0.8699	0.7781	2.0268	2.1901	0.0280	0.0799	0.8952	0.8799	0.7946	1.9321	2.5204	0.0232	0.0726	0.9064
MTF-GLP-FS	0.8791	0.7859	1.9797	2.1739	0.0449	0.0817	0.8776	0.8798	0.7895	1.9480	2.5129	0.0395	0.0845	0.8799
MTF-GLP-HPM-H	0.6794	0.6809	141.91	2.8846	0.0998	0.1255	0.7897	0.8875	0.7973	1.9499	2.7653	0.0970	0.1304	0.7863
BIEDN	0.9284	0.8676	1.4321	1.5748	0.0329	**0.0387**	0.9296	0.9284	0.8666	1.4303	1.8386	0.0325	**0.0403**	0.9285
ArbRPN (ours)	**0.9363**	**0.8822**	**1.2478**	**1.3678**	0.0229	0.0456	**0.9331**	**0.9324**	**0.8745**	**1.2731**	**1.6443**	0.0227	0.0438	0.9353
	Average							Standard deviation						
EXP	0.7502	0.5416	2.4271	2.9965	**0.0000**	0.1073	0.8923	0.0101	0.0133	1.2623	0.4529	**0.0000**	0.0040	0.0048
PCA	0.7530	0.6195	2.3299	3.0201	0.0423	0.1881	0.8014	0.0449	0.0476	1.1510	0.3850	0.0105	0.0541	0.0403
IHS	0.7442	0.5962	2.3196	3.1805	0.0963	0.2067	0.7361	0.0363	0.0313	1.1653	0.4153	0.0032	0.0387	0.0164
BDSD	0.8588	0.7477	1.8757	2.2679	0.0226	0.0754	0.9229	0.0207	0.0665	0.8267	0.2542	0.0054	0.0407	0.0193
MTF-GLP	0.8739	0.7646	1.7991	2.3611	0.0783	0.1223	0.8111	0.0049	0.0082	0.6908	0.3754	0.0079	0.0021	0.0047
C-GSA	0.8170	0.7106	2.7698	2.5053	0.1003	0.2022	0.7484	0.0909	0.0982	2.4512	**0.1521**	0.0358	0.0671	0.0570
BDSD-PC	0.8629	0.7472	1.8615	2.2025	0.0260	0.0837	0.8980	0.0196	0.0612	0.8170	0.3119	0.0025	0.0112	0.0074
MTF-GLP-FS	0.8814	0.7867	1.6884	2.1712	0.0443	0.0810	0.8780	0.0022	**0.0019**	0.6033	0.3431	0.0045	0.0028	0.0017
MTF-GLP-HPM-H	0.6034	0.6618	185.57	2.6935	0.1050	0.1254	0.7862	0.2750	0.1245	282.67	0.2353	0.0115	0.0069	0.0036
BIEDN	0.9290	0.8691	1.3054	1.5788	0.0307	**0.0409**	0.9295	**0.0007**	0.0040	0.2937	0.2579	0.0035	**0.0023**	**0.0010**
ArbRPN (ours)	**0.9404**	**0.8881**	**1.1496**	**1.3865**	0.0240	0.0449	**0.9312**	0.0119	0.0201	**0.1836**	0.2489	0.0020	0.0045	0.0053

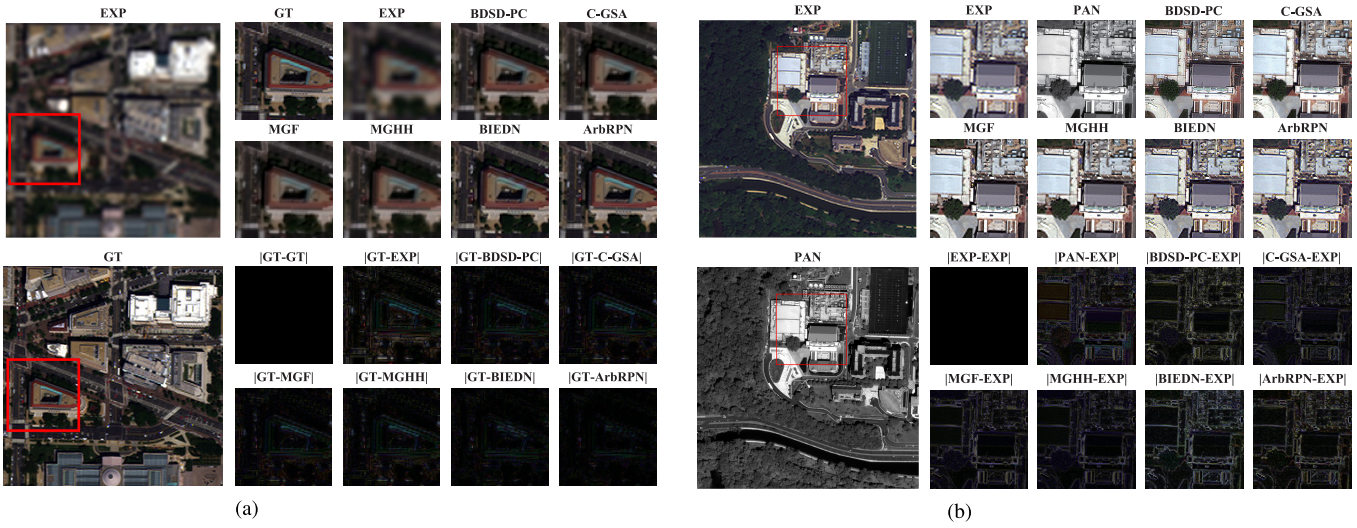


Fig. 9. Comparison of results obtained by models for MS images with various numbers of bands acquired by the WV2 sensor. MGF refers to MTF-GLP-FS. MGHH refers to MTF-GLP-HPM-H. (a) Results at RR. (b) Results at FR.

achieving competitive spectral preservation. A comparison of visual effects for WV2 data is shown in Fig. 9, both at reduced and FR. Similar to the results in the QB test case, our ArbRPN also achieves a better tradeoff between spectral preservation and details enhancement than the benchmark.

3) Study of Model Efficiency: As shown in Table IV, the parameters of the proposed ArbRPN are roughly similar to that of the TFNET and fewer than that of the BIEDN. Our ArbRPN achieves better results than the compared networks for MS images with a fixed number of bands. Concerning models for

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 12:30:08 UTC from IEEE Xplore. Restrictions apply.

TABLE III

RESULTS OF THE COMPARED METHODS ON MS IMAGES WITH DIFFERENT NUMBERS OF BANDS FROM WV2 TEST SETS. '-' INDICATES THAT THERE IS NO RESULT CORRESPONDING TO THE METRIC DUE TO INVALID COMPUTATION. THE BEST RESULTS ARE IN BOLDFACE, WHILE THE SECOND-BEST RESULTS ARE UNDERLINED

Methods	RR Test				FR Test			RR Test				FR Test		
	CC↑	Q2n↑	ERGAS↓	SAM↓	$D_\lambda\downarrow$	$D_s\downarrow$	QNR↑	CC↑	Q2n↑	ERGAS↓	SAM↓	$D_\lambda\downarrow$	$D_s\downarrow$	QNR↑
	FIX-1							FIX-2						
EXP	0.7723	0.5824	7.7002	-	-	0.0528	-	0.7758	0.5870	7.4767	5.0041	**0.0000**	0.0629	**0.9371**
PCA	0.7406	0.5840	7.6734	-	-	0.3276	-	0.7521	0.6355	7.3497	6.4856	0.0881	0.2298	0.7046
IHS	0.7406	0.5840	7.6734	-	-	0.3276	-	0.8294	0.6721	6.8276	5.4450	0.0748	0.2224	0.7238
BDSD	0.8429	0.6691	6.7036	-	-	0.1453	-	0.8742	0.7810	5.5566	5.0054	0.0384	0.0615	0.9022
MTF-GLP	0.8867	0.7857	5.5135	-	-	0.1334	-	0.8885	0.7971	5.3548	4.9142	0.0993	0.1339	0.7824
C-GSA	0.7406	0.6116	17.398	-	-	0.1729	-	0.8681	0.7829	5.7352	5.8511	0.0598	0.0916	0.8554
BDSD-PC	0.8474	0.6818	6.5958	-	-	0.0953	-	0.8809	0.7826	5.4786	4.8782	0.0476	**0.0424**	0.9140
MTF-GLP-FS	0.8955	0.7971	5.1913	-	-	0.0768	-	0.8983	0.8129	5.0586	4.8478	0.0788	0.0790	0.8503
MTF-GLP-HPM-H	0.3847	0.5939	714.94	-	-	0.1013	-	0.7430	0.7536	154.84	5.8022	0.0759	0.0659	0.8648
BIEDN	**0.9605**	**0.9214**	**3.0818**	-	-	**0.0400**	-	0.9608	**0.9220**	2.9829	3.0958	0.0943	0.0435	0.8681
ArbRPN (ours)	0.9538	0.9107	3.3660	-	-	0.0497	-	**0.9619**	0.9216	**2.9290**	**3.0109**	0.0607	0.0440	0.8999
	FIX-3							FIX-4						
EXP	0.7784	0.5896	7.6064	6.2844	**0.0000**	0.0605	**0.9395**	0.7763	0.5857	7.4524	6.9067	**0.0000**	0.0605	**0.9395**
PCA	0.7672	0.6670	7.2768	7.9049	0.0702	0.1891	0.7560	0.7974	0.6849	6.7292	8.1683	0.0683	0.1641	0.7812
IHS	0.8583	0.7081	6.5911	6.7252	0.0857	0.1756	0.7561	0.8663	0.7271	6.2362	7.2695	0.0811	0.1650	0.7691
BDSD	0.8936	0.8212	5.1495	6.2066	0.0392	0.0463	0.9164	0.8959	0.8235	4.9297	6.6952	0.0493	0.0419	0.9110
MTF-GLP	0.8916	0.8023	5.3755	6.0949	0.0895	0.1284	0.7955	0.8907	0.7996	5.2511	6.6617	0.0961	0.1331	0.7855
C-GSA	0.8911	0.8070	5.2954	6.1670	0.0572	0.1003	0.8496	0.8967	0.8124	5.0333	6.5826	0.0598	0.0916	0.8554
BDSD-PC	0.8983	0.8245	5.0617	6.0563	0.0431	0.0579	0.9031	0.9045	0.8333	4.7407	6.4607	0.0476	0.0424	0.9140
MTF-GLP-FS	0.9022	0.8202	5.0431	6.0473	0.0738	0.0783	0.8554	0.9017	0.8189	4.9171	6.5973	0.0788	0.0790	0.8503
MTF-GLP-HPM-H	0.8676	0.8236	11.333	5.5917	0.0917	0.0903	0.8278	0.8858	0.8221	16.552	6.1686	0.0759	0.0659	0.8648
BIEDN	0.9629	0.9221	2.9905	3.8629	0.0806	0.0458	0.8785	0.9621	0.9199	2.9075	4.2228	0.0853	0.0474	0.8728
ArbRPN (ours)	**0.9678**	**0.9278**	**2.7589**	**3.5619**	0.0466	**0.0342**	0.9218	**0.9695**	**0.9293**	**2.5994**	**3.8074**	0.0488	**0.0336**	0.9207
	FIX-5							FIX-6						
EXP	0.7787	0.5863	7.5777	7.3407	**0.0000**	0.0610	**0.9390**	0.7773	0.5883	7.6035	7.3950	**0.0000**	0.0607	**0.9393**
PCA	0.7357	0.6780	7.2885	8.8682	0.0662	0.1721	0.7744	0.7796	0.6760	7.0437	8.8474	0.0606	0.1607	0.7901
IHS	0.8682	0.7299	6.2998	7.6973	0.0845	0.1525	0.7775	0.8724	0.7304	6.2376	7.7060	0.0807	0.1497	0.7837
BDSD	0.8949	0.8242	5.1119	7.2192	0.0535	0.0457	0.9033	0.8966	0.8285	5.0357	7.2565	0.0553	0.0543	0.8935
MTF-GLP	0.8914	0.8000	5.3771	7.0936	0.0965	0.1350	0.7826	0.8922	0.8042	5.3392	7.1118	0.0919	0.1325	0.7891
C-GSA	0.8987	0.8153	5.1740	7.1227	0.0630	0.0958	0.8483	0.9007	0.8209	5.0863	7.0695	0.0586	0.0911	0.8569
BDSD-PC	0.9048	0.8344	4.8893	6.9163	0.0483	0.0412	0.9139	0.9074	0.8404	4.7871	6.9238	0.0473	0.0385	0.9175
MTF-GLP-FS	0.9021	0.8193	5.0616	7.0702	0.0793	0.0827	0.8457	0.9027	0.8238	5.0019	7.0521	0.0764	0.0807	0.8504
MTF-GLP-HPM-H	0.9098	0.8229	4.8840	6.4877	0.1050	0.0977	0.8091	0.9092	0.8273	4.9156	6.5202	0.1010	0.0943	0.8163
BIEDN	0.9621	0.9199	3.0228	4.5391	0.0835	0.0484	0.8732	0.9631	0.9230	2.9725	4.5995	0.0799	0.0461	0.8789
ArbRPN (ours)	**0.9704**	**0.9318**	**2.6359**	**4.0212**	0.0421	**0.0312**	0.9289	**0.9732**	**0.9380**	**2.5188**	**3.9797**	0.0394	**0.0285**	0.9340
	FIX-7							FIX-8						
EXP	0.7784	0.5880	7.6127	7.7760	**0.0000**	0.0618	**0.9382**	0.7925	0.6080	7.4350	7.7174	**0.0000**	0.0593	**0.9407**
PCA	0.7514	0.6639	7.3038	9.5519	0.0597	0.1540	0.7970	0.7649	0.6674	7.2024	9.5091	0.0582	0.1511	0.8004
IHS	0.8731	0.7266	6.2748	8.0832	0.0789	0.1331	0.8014	0.8777	0.7338	6.1000	7.9746	0.0842	0.1286	0.7996
BDSD	0.8947	0.8258	5.1305	7.6603	0.0597	0.0673	0.8771	0.8898	0.8156	5.2645	8.0761	0.0821	0.1049	0.8221
MTF-GLP	0.8921	0.8037	5.3861	7.4684	0.0931	0.1333	0.7874	0.9024	0.8179	5.1425	7.3228	0.0884	0.1501	0.7758
C-GSA	0.9068	0.8228	5.1121	7.4558	0.0625	0.0943	0.8501	0.9018	0.8230	5.1099	7.5766	0.0639	0.0944	0.8488
BDSD-PC	0.9006	0.8390	4.8504	7.2579	0.0468	0.0411	0.9158	0.9069	0.8383	4.8512	7.3370	0.0466	0.0414	0.9156
MTF-GLP-FS	0.9024	0.8229	5.0676	7.4351	0.0773	0.0823	0.8481	0.9020	0.8222	5.0740	7.5252	0.0782	0.0823	0.8472
MTF-GLP-HPM-H	0.9127	0.8261	4.8655	6.8628	0.1024	0.0972	0.8125	0.9107	0.8249	4.8906	6.9665	0.1037	0.0973	0.8112
BIEDN	0.9627	0.9224	3.0136	4.8019	0.0817	0.0462	0.8773	0.9621	0.9211	3.0238	4.8966	0.0823	0.0467	0.8761
ArbRPN (ours)	**0.9726**	**0.9356**	**2.5526**	**4.1465**	0.0419	**0.0276**	0.9326	**0.9739**	**0.9386**	**2.4871**	**4.1316**	0.0410	**0.0267**	0.9344
	Average							Standard deviation						
EXP	0.7787	0.5894	7.5581	6.9178	**0.0000**	0.0599	**0.9390**	0.0059	0.0078	0.0933	0.9871	**0.0000**	0.0031	**0.0011**
PCA	0.7611	0.6571	7.2334	8.4765	0.0673	0.1936	0.7720	0.0205	0.0330	0.2694	1.0723	0.0102	0.0598	0.0333
IHS	0.8483	0.7015	6.5301	7.2715	0.0814	0.1818	0.7730	0.0460	0.0516	0.5175	0.9272	0.0038	0.0658	0.0270
BDSD	0.8853	0.7986	5.3603	6.8742	0.0539	0.0709	0.8894	0.0186	0.0545	0.5734	1.0245	0.0148	0.0361	0.0323
MTF-GLP	0.8920	0.8013	5.3425	6.6668	0.0935	0.1350	0.7855	0.0046	0.0089	0.1082	0.8992	0.0040	0.0064	0.0062
C-GSA	0.8756	0.7870	6.7430	6.8322	0.0607	0.0942	0.8521	0.0558	0.0721	4.3111	0.6520	0.0025	0.0032	0.0037
BDSD-PC	0.8939	0.8093	5.1568	6.5472	0.0468	0.0436	0.9134	0.0206	0.0549	0.6270	0.8603	0.0017	0.0065	0.0047
MTF-GLP-FS	0.9009	0.8172	5.0519	6.6536	0.0775	0.0806	0.8496	0.0026	0.0088	0.0766	0.9429	0.0019	**0.0019**	0.0031
MTF-GLP-HPM-H	0.8154	0.7868	114.65	6.3428	0.0937	0.0869	0.8295	0.1831	0.0818	247.97	0.5162	0.0129	0.0146	0.0249
BIEDN	0.9620	0.9215	2.9994	4.2884	0.0839	0.0455	0.8750	**0.0009**	**0.0011**	**0.0502**	0.6326	0.0049	0.0026	0.0039
ArbRPN (ours)	**0.9679**	**0.9292**	**2.7310**	**3.8085**	0.0458	**0.0344**	0.9246	0.0069	0.0094	0.2942	**0.4062**	0.0074	0.0083	0.0122

MS images with any number of bands, our ArbRPN obtains better results than the BIEDN with fewer parameters. These observations demonstrate that our ArbRPN is more efficient from a number of parameters point of view than the compared DL-based methods. As for the inference time, all the results are tested on a desktop with a 1060 GPU for the reconstruction of a 256×256 MS image. Our ArbRPN has a comparative performance than the BIEDN with regard to this point. It is worth mentioning that although the inference times of our ArbRPN and the one of the BIEDN are longer than the other DL-based methods, both methods can achieve near real-time processing for MS images of size 256×256 on desktops with a consumer-grade GPU.

C. Ablation Study

1) Effectiveness of the Key Modules in ArbRPN: In this section, we conducted several comparative experiments to validate the effectiveness of the proposed modules. In this section, the number of the HUs is set to 1 and the number

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 12:30:08 UTC from IEEE Xplore. Restrictions apply.

TABLE IV

RUNNING TIME AND PARAMETERS OF DIFFERENT METHODS. '-' REFERS TO NO LEARNABLE PARAMETER IN THE CORRESPONDING METHOD. THE RUNNING TIME IS TESTED ON A DESKTOP WITH A CPU OF INTEL CORE I5-4460 AND A GPU OF NVIDIA GTX-1060. TRADITIONAL METHODS ARE TESTED USING CPU, WHILE DL-BASED ARE TESTED USING GPU

	Methods	Time(s)	#Params.
Traditional methods	EXP	0.0167	-
	PCA	0.0167	-
	IHS	0.0026	-
	BDSD	0.0370	-
	MTF-GLP	0.0542	-
	C-GSA	0.3210	-
	BDSD-PC	0.0639	-
	MTF-GLP-FS	0.0583	-
	MTF-GLP-HPM-H	0.0649	-
DL-based methods	PNN	0.0012	80K
	DRPNN	0.0028	1.64M
	PANNET	0.0031	77K
	DICNN1	**0.0008**	**42K**
	MSDCNN	0.0019	190K
	TFNET	0.0047	2.36M
	BIEDN	0.0431	3.8M
	ArbRPN (ours)	0.0490	2.34M

TABLE V

RESULTS OF THE ABLATION STUDY. PERFORMANCE IS EVALUATED ON THE TEST SET OF FIX-4 ACQUIRED BY THE QB SENSOR. THE BEST RESULTS ARE IN BOLDFACE

Methods	RR Test				FR Test	#Params.
	CC↑	Q2n ↑	EGRAS ↓	SAM ↓	QNR↑	
RNN	0.9230	0.8584	1.4825	1.8501	0.8947	**453.2K**
RNN+RL	0.9309	0.8714	1.3954	1.7461	0.9119	**453.2K**
BiRNN+RL	0.9339	0.8759	1.3883	1.7449	**0.9405**	461.5K
BiRNN+RL+PJ(ArbRPN)	**0.9395**	**0.8861**	**1.3374**	**1.7040**	0.9397	469.7K
ArbRPN-SD	0.9252	0.8619	1.4808	1.8661	0.9207	461.5K

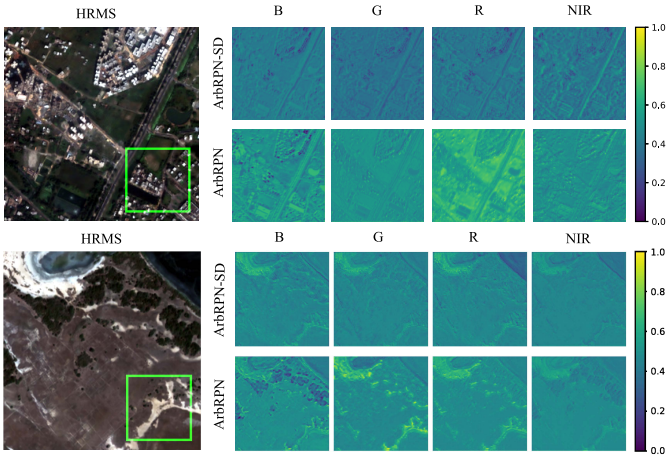


Fig. 10. Comparison of the learned features for the reconstruction bands using the ArbRPN-SD and the ArbRPN. The first sample is from the output of the FU, while the second is from the output of the BU.

TABLE VI

PERFORMANCE COMPARISON BETWEEN THE VANILLA BIRNN AND THE PROPOSED BIRNN. THE BEST RESULTS ARE IN BOLDFACE

Methods	RR Test				FR Test	#Params.
	CC↑	Q2n ↑	EGRAS ↓	SAM ↓	QNR↑	
Vanilla-BiRNN+RL	0.7935	0.1983	5.4821	8.6235	0.6641	469.8K
our-BiRNN+RL	**0.9339**	**0.8759**	**1.3883**	**1.7449**	**0.9405**	**461.5K**

TABLE VII

PERFORMANCE (EVALUATED THROUGH THE CC INDEX AT RR) OF THE DIFFERENT TRAINING METHODS ON THE QB TEST SETS. THE BEST RESULTS ARE IN BOLDFACE

Methods	FIX-1	FIX-2	FIX-3	FIX-4	Ave.	Std.
Fix-training	0.8583	0.8542	0.8572	0.9608	0.8826	0.0521
Alternative-training (Epoch)	**0.9301**	0.9174	0.9254	**0.9291**	0.9255	0.0058
Alternative-training (Iteration)	0.9276	0.9182	0.9255	0.9273	0.9247	0.0044
Mask-training	0.9228	**0.9280**	**0.9309**	0.9284	**0.9275**	**0.0034**

of the RBs in the FU/BU is set 3 for saving training time. The baseline of the ArbRPN is the RNN, which has the same network depth as that of the ArbRPN. Compared with the proposed ArbRPN, the RNN does not present the backward recurrence and RL. Besides, the RNN only injects the PAN features at the first recurrence by adding the PAN features only into the initial hidden state. According to the first property, the global and local RL are added to the RNN, thus achieving the RNN + RL. As shown in Table V, it is straightforward that the RL improves the performance. Then, to verify the effectiveness of the BiRNN, we replace the RNN with the proposed BiRNN architecture obtaining the BiRNN + RL. As shown in Table V, the architecture of the BiRNN also achieves an improvement in performance. This is because the BiRNN can embed the spectral dependency among all the bands from two directions facilitating the full exploration of the spectral dependency, while RNN only embeds the spectral dependency among the current band and the previous bands. Next, to validate the effectiveness of injecting PAN images into every band of the HRMS image, the features extracted from the PAN image are explicitly integrated into every hidden state in the BiRNN + RL to achieve the proposed ArbRPN. As shown in Table V, the injection of the PAN image into every band also achieves a performance improvement. This is because there is no long-term memory in the proposed ArbRPN. If the information of the PAN image is only fed into the network through the initial hidden state, this information will be forgotten in the subsequent recurrences.

2) Effectiveness of Learning Spectral Dependency: To show the significance of the spectral dependency for pansharpening, we remove the connections between adjacent forward recurrences and the connections between adjacent backward recurrences obtaining a new network named ArbRPN without spectral dependency (ArbRPN-SD), see Table V. In this way, the information from the previous forward/backward recurrence cannot be transferred to the current recurrence breaking the ability to capture the spectral dependency. As shown in Table V, ArbRPN-SD suffers from a considerable performance drop and only achieves a result that is slightly better than the RNN, thus demonstrating the significance of the spectral dependency. To further explore how the learned spectral dependency affects the details to be injected, we show the features extracted from the reconstructed bands using the ArbRPN and the ArbRPN-SD approaches. As shown in Fig. 10, it is clear that the extracted features for the different bands of the ArbRPN-SD are similar to each other demonstrating that the ArbRPN-SD has a reduced ability to inject band-dependent details. On the other hand, the extracted features by the proposed ArbRPN are clearly band-dependent.

3) Vanilla BiRNN Versus Our BiRNN: As mentioned in Section III-A, the FU and the BU are sequentially connected in our ArbRPN, differently from the parallel architecture in

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 12:30:08 UTC from IEEE Xplore. Restrictions apply.

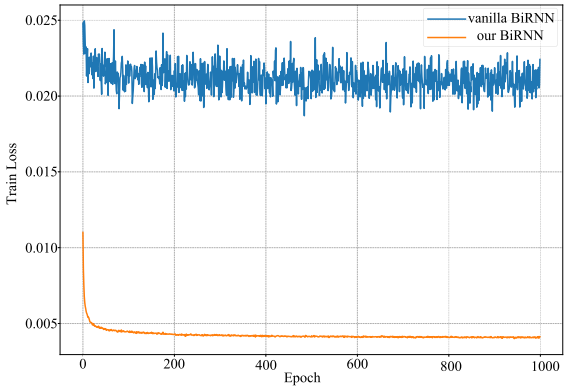


Fig. 11. Loss and CC curves of the vanilla BiRNN and our sequential BiRNN.

TABLE VIII

CROSS VALIDATION ON DIFFERENT DATASETS. THE PERFORMANCE IS EVALUATED ON THE QB TEST SET USING THE CC PERFORMANCE METRIC AT RR. THE BEST RESULTS ARE IN BOLDFACE

Training	Test					
	FIX-1	FIX-2	FIX-3	FIX-4	Ave.	Std.
FIX-1	**0.9256**	0.8746	0.8610	0.8644	0.8814	0.0300
MIX-2	0.9189	**0.9205**	0.9187	0.9181	0.9191	**0.0010**
MIX-3	0.9193	0.9196	0.9243	0.9244	**0.9219**	0.0028
MIX-4	0.9170	0.9179	**0.9247**	**0.9271**	0.9217	0.0050

the vanilla BiRNN. The reason for sequentially cascading the FU and the BU is that it can increase the network depth while keeping similar parameters with respect to the vanilla BiRNN. The quantitative results are shown in Table VI. The proposed BiRNN architecture achieves better performance than the vanilla BiRNN, which demonstrates the effectiveness of the sequential BiRNN architecture in the proposed ArbRPN. To explore the reason for the worse performance of the vanilla BiRN, we show the training loss curves of the vanilla-BiRNN + RL and our BiRNN + RL. Fig. 11 shows that the vanilla BiRNN is unstable and, moreover, it cannot achieve convergence during the training stage resulting in lower performance.

4) Effectiveness of Mask-Training: To demonstrate the effectiveness of the mask-training, we compare it with three other training methods. The first one is the training of the network only with images with a fixed number of bands, named fix-training. The second method, named alternative-training-epoch, alternatively trains the network with different FIX-N datasets in different epochs. The third one, named alternative-training-iteration, alternatively trains the network with different FIX-N datasets in different iterations. The performance of these training methods is reported in Table VII. We can observe that the network using the mask-training obtains the best overall (ave.) results. Moreover, the standard deviation (std.) of the results obtained by the mask-training is smaller than the fix-training and the alternative-training, which indicates that the mask-training is able to support the robustness of the proposed ArbRPN.

5) Performance Versus Model's Capacity: To investigate the performance against the capacity of the ArbRPN, we study two key hyper-parameters, i.e., the number of HUs marked as N and the number of RBs marked as L. First, we change N from 1 to 6 with a unitary step and L is fixed to 1. The performance of the six networks is shown in Fig. 12, from which we can find that the ArbRPN performance improves by incrementing N. Then, we change L from 1 to 6 with a unitary step and N is fixed to 1. Akin to the results obtained on N, the higher the value of L, the better the performance. To keep the number of parameters roughly similar to the TFNET [49] for a fair comparison, $N = 5$ and $L = 3$ are selected as the final settings for the proposed ArbRPN.

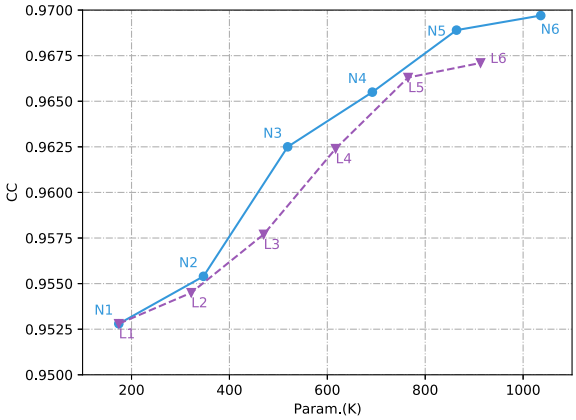


Fig. 12. Performance (using the CC metric at RR) versus parameters. N is the number of the HU in the proposed network. L is the number of RBs in the HU/FU.

D. Generalization on Band Numbers

To analyze the generalization of the proposed ArbRPN on MS images with different numbers of bands, we conduct several validation experiments by changing the training set of the network. As shown in Table VIII, we train four networks with four different training sets, i.e., MIX-1–MIX-4 (where FIX-1 is equal to MIX-1). The results of the four networks are shown in Table VIII. From these results, we can note two phenomenona as follow.

1) Observing the results along the row direction, we find that the ArbRPN achieves a great generalization on unseen samples. For example, the network trained with the MIX-2, which does not see samples with three or more bands in the training stage, still achieves favorable performance on FIX-3 and FIX-4 datasets. It is also true for the network trained with the MIX-3. As for the network trained with the FIX-1, it achieves worse results on MS images with more bands. This is because only one band can be used to train the proposed ArbRPN, making impossible for our ArbRPN to learn the spectral dependency among different bands. These results also indicate the significance of incorporating the spectral dependency for pansharpening.
2) Observing the results along the column direction, especially for the results on the FIX-3 and the FIX-4, we find that the performance of the network improves by considering a higher variety of types of MS images. This is because training with more types of MS images

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 12:30:08 UTC from IEEE Xplore. Restrictions apply.

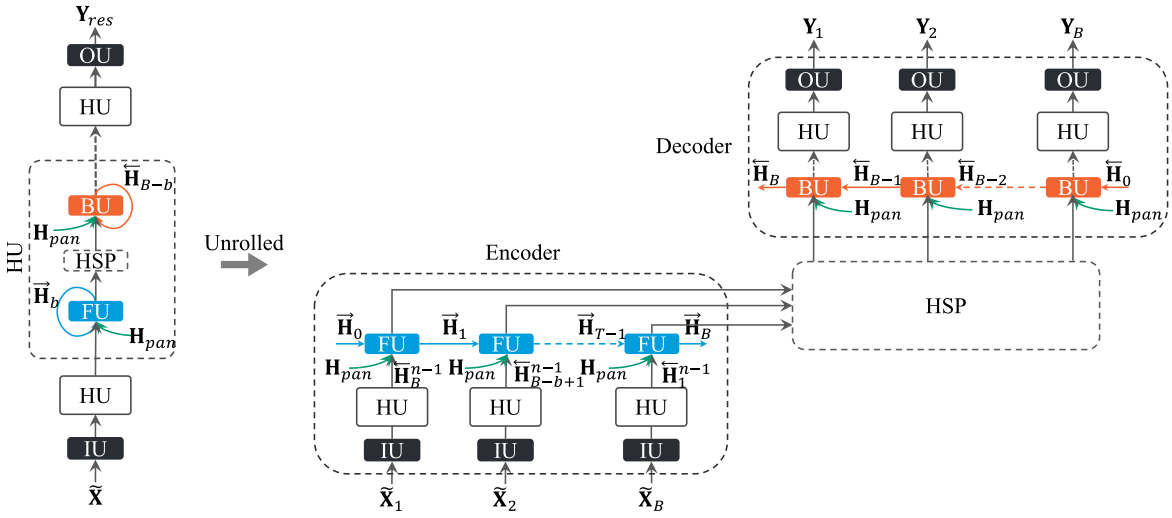


Fig. 13. Illustration of the proposed ArbRPN in a Seq2Seq manner. HSP refers to the hidden state pool (HSP), which is used to store hidden states.

(showing different numbers of bands) enables our ArbRPN to fully explore the spectral dependency among different bands.

V. Conclusion

In this article, we proposed a pansharpening network, named ArbRPN, to reconstruct HRMS images with arbitrary numbers of bands by using a single model. To process MS images with any number of bands, our ArbRPN dynamically changed the number of recurrence according to the number of bands in the LRMS image. RL was adopted to ensure that the majority of the spectral information of a reconstructed band was originated from the corresponding band of the LRMS image. Besides, a sequential BiRNN structure was devised to fully utilize the spectral dependency in the MS spectral bands. To overcome the forgetting of PAN features and to restore more spatial details, the information of the PAN image was explicitly injected into each band of the reconstructed MS image. In the training stage, we introduced mask-training to train our ArbRPN using mini-batches consisting of LRMS images with different numbers of bands improving the performance and the robustness of the proposed approach. In the ablation study, we verified the effectiveness of each module of the proposed architecture. Moreover, the spectral dependency was shown to be a key factor for getting a superior performance. The effectiveness of the proposed mask-training was also validated by comparing it with other training methods. Furthermore, an experiment was conducted to show the generalization ability of the proposed approach on MS images with different numbers of bands.

Appendix A
Relationship Between Our ArbRPN and ConvSeq2Seq Model

Our ArbRPN can also be viewed as a special case of the Seq2Seq model [55], which cannot be achieved by the vanilla BiRNN. Taking the connection between the FU and the BU in any HU as a dividing line, we split our ArbRPN into two parts, as shown in Fig. 13. The left part can be viewed as the encoder in the Seq2Seq and the right part can be viewed as the decoder in the Seq2Seq. The band embedding is stored in the HSP. The encoder is used to extract band features and to embed the spectral dependence into the hidden states, while the decoder is used to reconstruct bands of the HRMS image. The difference between the proposed BiRNN and the Seq2Seq model is that the proposed model can be viewed as a self-ensemble of numbers of Seq2Seq models by dividing the encoder and the decoder in any HU.

Appendix B
Ability of ArbRPN to Capture Spectral Dependency

The proposed ArbRPN can be regarded as an end-to-end network to learn the combination of two functions: one related to the generation of the spectral dependency and, the other one, about the details extraction. The above two functions can be implemented by two sub-networks. The first sub-network is utilized to map the inputs (arbitrary bands) and the PAN image to a variable representing the spectral dependency of the input bands. Hence, we have

$$\begin{aligned}\mathbf{Z} &= f_{\text{net}1}\left(\widetilde{\mathbf{X}}, \mathbf{X}_{\text{pan}}\right) \\ &= f_{\text{net}1}\left(\left[\widetilde{\mathbf{X}}_1, \mathbf{X}_{\text{pan}}\right], \left[\widetilde{\mathbf{X}}_2, \mathbf{X}_{\text{pan}}\right], \ldots, \left[\widetilde{\mathbf{X}}_b, \mathbf{X}_{\text{pan}}\right]\right) \quad (14)\end{aligned}$$

where $\widetilde{\mathbf{X}}$ refers to the upsampled MS image, $\widetilde{\mathbf{X}}_b$ is the bth band in the upsampled MS image $\widetilde{\mathbf{X}}$, $\mathbf{X}_{\text{pan}}$ is the PAN image, $f_{\text{net}1}(\cdot)$ is the sub-network to obtain the spectral dependency according to the arbitrary input bands and the PAN image. To capture spectral dependency from arbitrary input bands, the sub-network needs to be input-order-aware because the obtained spectral dependency should be flipped if the band order is flipped, which means different band orders resulting in different outcomes, i.e., assuming two bands in the MS image, we have that

$$f_{\text{net}1}\left(\left[\widetilde{\mathbf{X}}_2, \mathbf{X}_{\text{pan}}\right], \left[\widetilde{\mathbf{X}}_1, \mathbf{X}_{\text{pan}}\right]\right) \neq f_{\text{net}1}\left(\left[\widetilde{\mathbf{X}}_1, \mathbf{X}_{\text{pan}}\right], \left[\widetilde{\mathbf{X}}_2, \mathbf{X}_{\text{pan}}\right]\right). \quad (15)$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 12:30:08 UTC from IEEE Xplore. Restrictions apply.

Thus, the sub-network will map the input bands with different orders to different spectral dependencies. Hence, our ArbRPN has the character of awareness of the input order, because it is built starting from an RNN that is sensitive to the order of the inputs. According to (14), the spectral dependency represented by the variable $\mathbf{Z}$ not only depends on the input bands and the PAN image but also on the band order in $\widetilde{\mathbf{X}}$. After obtaining the spectral dependency according to the MS image, a sub-network is further used to extract details according to the obtained spectral dependency $\mathbf{Z}$, the input bands, and the PAN image. Hence, we have

$$\begin{aligned}\mathbf{Y}_{\text{res}} &= f_{\text{net2}}\left(\widetilde{\mathbf{X}}, \mathbf{X}_{\text{pan}}, \mathbf{Z}\right) \\ &= f_{\text{net2}}\left(\left[\widetilde{\mathbf{X}}_1, \mathbf{X}_{\text{pan}}\right], \left[\widetilde{\mathbf{X}}_2, \mathbf{X}_{\text{pan}}\right], \ldots, \left[\widetilde{\mathbf{X}}_b, \mathbf{X}_{\text{pan}}\right], \mathbf{Z}\right)\end{aligned} \tag{16}$$

where $\mathbf{Z}$ is obtained by (14), $\mathbf{Y}_{\text{res}}$ are the generated details to be added to the upsampled LRMS input, $f_{\text{net2}}(\cdot)$ is the sub-network related to the generation of details using the MS image, the PAN image, and the spectral dependency. Observing (14) and (16), we can find that they have the same inputs, except for $\mathbf{Z}$. Moreover, $\mathbf{Z}$ can be obtained by (14), which means that the final extracted details are related to the same inputs, i.e., the input MS image and the PAN image. Therefore, we can combine the two sub-networks, i.e., $f_{net1}(\cdot)$ and $f_{net1}(\cdot)$, to build an end-to-end network, i.e., the proposed ArbRPN, which implicitly integrates the two above-mentioned steps into an end-to-end network. In our early experiments, we used the explicit way to implement the two steps, i.e., (14) and (16). Specifically, we exploited the RNN attention to obtain an explicit spectral dependency. It is worth to be noted that attention can calculate the correlation of features, thus, this mechanism can naturally be utilized to represent the spectral dependency. As shown in Fig. 13, our ArbRPN can be regarded as a special Seq2Seq model. In our early experiments, we used a Seq2Seq model only, including FU in HU, which means that the BU in Fig. 13 is replaced by the FU in the model. The encoder embeds the input bands into a series of embedding tensors. We utilized the attention mechanism to capture spectral dependency from the embedding tensors saved in the HSP. Afterward, the dependency for arbitrary input bands has explicitly been determined by the attention mechanism. However, this approach has not been used in this article because: 1) the performance improvement is paid by a relevant increase of the computation burden (especially for an elevated number of spectral bands to be fused) and 2) the exploration of advanced RNNs or other DL technologies is beyond the scope of this article, mainly based on the presentation of a solution to process MS images with any bands number using a single model.

REFERENCES

[1] DigitalGlobe. (2013). *Spectral Response for Digital Globe Earth Imaging Instruments*. [Online]. Available: https://dg-cms-uploads-production.s3.amazonaws.com/uploads/document/file/105/DigitalGlobe_Spectral_Response_1.pdf

[2] C. Liu *et al.*, "Band-independent encoder–decoder network for pan-sharpening of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 5208–5223, Jul. 2020.

[3] W. Huang, L. Xiao, Z. Wei, H. Liu, and S. Tang, "A new pan-sharpening method with deep neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 5, pp. 1037–1041, May 2015.

[4] L. Medsker and L. C. Jain, *Recurrent Neural Networks: Design and Applications*. Boca Raton, FL, USA: CRC Press, 1999.

[5] R. Haydn, "Application of the IHS color transform to the processing of multisensor data and image enhancement," in *Proc. Int. Symp. Remote Sens. Arid Semi-Arid Lands*, Cairo, Egypt, 1982.

[6] W. J. Carper, T. M. Lillesand, and R. W. Kiefer, "The use of intensity-hue-saturation transformations for merging SPOT panchromatic and multispectral image data," *Photogramm. Eng. Remote Sens.*, vol. 56, no. 4, pp. 459–467, Apr. 1990.

[7] A. R. Gillespie, A. B. Kahle, and R. E. Walker, "Color enhancement of highly correlated images. II. Channel ratio and 'chromaticity' transformation techniques," *Remote Sens. Environ.*, vol. 22, no. 3, pp. 343–365, 1987.

[8] C. A. Laben and B. V. Brower, "Process for enhancing the spatial resolution of multispectral imagery using pan-sharpening," U.S. Patent 6 011 875, Jan. 4, 2000.

[9] S. Rahmani, M. Strait, D. Merkurjev, M. Moeller, and T. Wittman, "An adaptive IHS pan-sharpening method," *IEEE Geosci. Remote Sens. Lett.*, vol. 7, no. 4, pp. 746–750, Oct. 2010.

[10] J. Choi, K. Yu, and Y. Kim, "A new adaptive component-substitution-based satellite image fusion by using partial replacement," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 1, pp. 295–309, Jan. 2011.

[11] Q. Xu, B. Li, Y. Zhang, and L. Ding, "High-fidelity component substitution pansharpening by the fitting of substitution data," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 11, pp. 7380–7392, Nov. 2014.

[12] G. Vivone, "Robust band-dependent spatial-detail approaches for panchromatic sharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6421–6433, Sep. 2019.

[13] G. Vivone, P. Addesso, R. Restaino, M. D. Mura, and J. Chanussot, "Pansharpening based on deconvolution for multiband filter estimation," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 540–553, Jan. 2019.

[14] B. Aiazzi, S. Baronti, and M. Selva, "Improving component substitution pansharpening through multivariate regression of MS +Pan data," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 10, pp. 3230–3239, Oct. 2007.

[15] B. Aiazzi, L. Alparone, S. Baronti, A. Garzelli, and M. Selva, "MTF-tailored multiscale fusion of high-resolution MS and PAN imagery," *Photogramm. Eng. Remote Sens.*, vol. 72, no. 5, pp. 591–596, May 2006.

[16] X. Otazu, M. González-Audícana, O. Fors, and J. Núñez, "Introduction of sensor spectral response into image fusion methods. Application to wavelet-based methods," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 10, pp. 2376–2385, Oct. 2005.

[17] V. P. Shah, N. H. Younan, and R. L. King, "An efficient pan-sharpening method via a combined adaptive PCA approach and contourlets," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 5, pp. 1323–1335, May 2008.

[18] T.-M. Tu, S.-C. Su, H.-C. Shyu, and P. S. Huang, "A new look at IHS-like image fusion methods," *Inf. Fusion*, vol. 2, no. 3, pp. 177–186, Sep. 2001.

[19] G. Vivone, R. Restaino, M. Dalla Mura, G. Licciardi, and J. Chanussot, "Contrast and error-based fusion schemes for multispectral image pansharpening," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 5, pp. 930–934, May 2014.

[20] L. Alparone, L. Wald, J. Chanussot, C. Thomas, P. Gamba, and L. M. Bruce, "Comparison of pansharpening algorithms: Outcome of the 2006 GRS-S data-fusion contest," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 10, pp. 3012–3021, Oct. 2007.

[21] R. Restaino, M. D. Mura, G. Vivone, and J. Chanussot, "Context-adaptive pansharpening based on image segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 2, pp. 753–766, Feb. 2017.

[22] G. Vivone, R. Restaino, and J. Chanussot, "Full scale regression-based injection coefficients for panchromatic sharpening," *IEEE Trans. Image Process.*, vol. 27, no. 7, pp. 3418–3431, Jul. 2018.

[23] G. Vivone, S. Marano, and J. Chanussot, "Pansharpening: Context-based generalized Laplacian pyramids by robust regression," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 9, pp. 6152–6167, Sep. 2020.

[24] S. Lolli, L. Alparone, A. Garzelli, and G. Vivone, "Haze correction for contrast-based multispectral pansharpening," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 12, pp. 2255–2259, Dec. 2017.

[25] C. Ballester, V. Caselles, L. Igual, J. Verdera, and B. Rougé, "A variational model for P+XS image fusion," *Int. J. Comput. Vis.*, vol. 69, no. 1, pp. 43–58, Aug. 2006.

[26] F. Fang, F. Li, C. Shen, and G. Zhang, "A variational approach for pan-sharpening," *IEEE Trans. Image Process.*, vol. 22, no. 7, pp. 2822–2834, Jul. 2013.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 12:30:08 UTC from IEEE Xplore. Restrictions apply.

[27] L. Deng, G. Vivone, W. Guo, M. D. Mura, and J. Chanussot, "A variational pansharpening approach based on reproducible kernel Hilbert space and heaviside function," *IEEE Trans. Image Process.*, vol. 27, no. 9, pp. 4330–4344, Sep. 2018.

[28] P. Liu, L. Xiao, and T. Li, "A variational pan-sharpening method based on spatial fractional-order geometry and spectral–spatial low-rank priors," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 3, pp. 1788–1802, Mar. 2018.

[29] X. Fu, Z. Lin, Y. Huang, and X. Ding, "A variational pan-sharpening with local gradient constraints," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Long Beach, CA, USA, Jun. 2019, pp. 10257–10266.

[30] N. D. A. Mascarenhas, G. J. F. Banon, and A. L. B. Candeias, "Multispectral image data fusion under a Bayesian approach," *Int. J. Remote Sens.*, vol. 17, no. 8, pp. 1457–1471, May 1996.

[31] D. Fasbender, J. Radoux, and P. Bogaert, "Bayesian data fusion for adaptable image pansharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 6, pp. 1847–1857, Jun. 2008.

[32] H. K. Zhang and B. Huang, "A new look at image fusion methods from a Bayesian perspective," *Remote Sens.*, vol. 7, no. 6, pp. 6828–6861, May 2015.

[33] P. Guo, P. Zhuang, and Y. Guo, "Bayesian pan-sharpening with multiorder gradient-based deep network constraints," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 950–962, Mar. 2020, doi: 10.1109/JSTARS.2020.2975000.

[34] S. Li and B. Yang, "A new pan-sharpening method using a compressed sensing technique," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 2, pp. 738–746, Feb. 2011.

[35] M. Ghahremani and H. Ghassemian, "Remote sensing image fusion using ripplet transform and compressed sensing," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 3, pp. 502–506, Mar. 2015.

[36] M. R. Vicinanza, R. Restaino, G. Vivone, M. Dalla Mura, and J. Chanussot, "A pansharpening method based on the sparse representation of injected details," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 1, pp. 180–184, Jan. 2015.

[37] S. Tang, N. Zhou, and L. Xiao, "Pansharpening via locality-constrained sparse representation," in *Proc. Brit. Mach. Vis. Conf.*, London, U.K., 2017, p. 180.

[38] H. Yin, "PAN-guided cross-resolution projection for local adaptive sparse representation-based pansharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4938–4950, Jul. 2019.

[39] R. Gogineni and A. Chaturvedi, "A robust pansharpening algorithm based on convolutional sparse coding for spatial enhancement," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 10, pp. 4024–4037, Oct. 2019.

[40] R. Fei, J. Zhang, J. Liu, F. Du, P. Chang, and J. Hu, "Convolutional sparse representation of injected details for pansharpening," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 10, pp. 1595–1599, Oct. 2019.

[41] C. Dong, C. C. Loy, K. He, and X. Tang, "Image super-resolution using deep convolutional networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 38, no. 2, pp. 295–307, Feb. 2015.

[42] G. Masi, D. Cozzolino, L. Verdoliva, and G. Scarpa, "Pansharpening by convolutional neural networks," *Remote Sens.*, vol. 8, no. 7, pp. 594–615, Jul. 2016.

[43] Y. Wei and Q. Yuan, "Deep residual learning for remote sensed imagery pansharpening," in *Proc. Int. Workshop Remote Sens. Intell. Process. (RSIP)*, Shanghai, China, May 2017.

[44] J. Yang, X. Fu, Y. Hu, Y. Huang, X. Ding, and J. Paisley, "PanNet: A deep network architecture for pan-sharpening," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Venice, Italy, Oct. 2017, pp. 5449–5457.

[45] Y. Zhang, C. Liu, M. Sun, and Y. Ou, "Pan-sharpening using an efficient bidirectional pyramid network," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5549–5563, Aug. 2019.

[46] Q. Yuan, Y. Wei, X. Meng, H. Shen, and L. Zhang, "A multiscale and multidepth convolutional neural network for remote sensing imagery pan-sharpening," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 3, pp. 978–989, Mar. 2018.

[47] G. Scarpa, S. Vitale, and D. Cozzolino, "Target-adaptive CNN-based pansharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 9, pp. 5443–5457, Sep. 2018.

[48] L. He *et al.*, "Pansharpening via detail injection based convolutional neural networks," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 4, pp. 1188–1204, Apr. 2019.

[49] X. Liu, Q. Liu, and Y. Wang, "Remote sensing image fusion based on two-stream fusion network," *Inf. Fusion*, vol. 55, pp. 1–15, Mar. 2020.

[50] Z.-C. Wu *et al.*, "A new variational approach based on proximal deep injection and gradient intensity similarity for spatio-spectral image fusion," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 6277–6290, Oct. 2020, doi: 10.1109/JSTARS.2020.3030129.

[51] X. Fu, W. Wang, Y. Huang, X. Ding, and J. Paisley, "Deep multiscale detail networks for multiband spectral image sharpening," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 5, pp. 2090–2104, May 2021.

[52] Z.-C. Wu, T.-Z. Huang, L.-J. Deng, J.-F. Hu, and G. Vivone, "VO+net: An adaptive approach using variational optimization and deep learning for panchromatic sharpening," *IEEE Trans. Geosci. Remote Sens.*, early access, Mar. 26, 2021, doi: 10.1109/TGRS.2021.3066425.

[53] L.-J. Deng, G. Vivone, C. Jin, and J. Chanussot, "Detail injection-based deep convolutional neural networks for pansharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 8, pp. 6995–7010, Aug. 2021.

[54] M. Schuster and K. K. Paliwal, "Bidirectional recurrent neural networks," *IEEE Trans. Signal Process.*, vol. 45, no. 11, pp. 2673–2681, Nov. 1997.

[55] J. Gehring, M. Auli, D. Grangier, D. Yarats, and Y. N. Dauphin, "Convolutional sequence to sequence learning," in *Proc. Int. Conf. Mach. Learn.*, Sydney, NSW, Australia, 2017, pp. 1243–1252.

[56] X. Shi, Z. Chen, H. Wang, D.-Y. Yeung, W.-K. Wong, and W.-C. Woo, "Convolutional LSTM network: A machine learning approach for precipitation nowcasting," in *Proc. Adv. Neural Inf. Process. Syst.*, Montreal, QC, Canada, 2015, pp. 802–810.

[57] N. Ballas, L. Yao, C. Pal, and A. Courville, "Delving deeper into convolutional networks for learning video representations," in *Proc. Int. Conf. Learn. Represent.*, San Juan, PR, USA, 2016, pp. 1–11.

[58] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Las Vegas, NV, USA, Jun. 2016, pp. 770–778.

[59] B. Lim, S. Son, H. Kim, S. Nah, and K. M. Lee, "Enhanced deep residual networks for single image super-resolution," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Honolulu, HI, USA, Jul. 2017, pp. 136–144.

[60] W. L. Taylor, "'Cloze procedure': A new tool for measuring readability," *Journalism Quart.*, vol. 30, no. 4, pp. 415–433, 1953.

[61] A. Lopez-del Rio, M. Martin, A. Perera-Lluna, and R. Saidi, "Effect of sequence padding on the performance of deep learning models in archaeal protein functional prediction," *Sci. Rep.*, vol. 10, no. 1, pp. 1–14, Dec. 2020.

[62] B. Aiazzi, L. Alparone, S. Baronti, and A. Garzelli, "Context-driven fusion of high spatial and spectral resolution images based on oversampled multiresolution analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 40, no. 10, pp. 2300–2312, Oct. 2002.

[63] G. Vivone *et al.*, "A critical comparison among pansharpening algorithms," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 5, pp. 2565–2586, May 2015.

[64] L. Wald, T. Ranchin, and M. Mangolini, "Fusion of satellite images of different spatial resolutions: Assessing the quality of resulting images," *Photogramm. Eng. Remote Sens.*, vol. 63, no. 6, pp. 691–699, 1997.

[65] G. Vivone *et al.*, "A new benchmark based on recent advances in multispectral pansharpening: Revisiting pansharpening with classical and emerging pansharpening methods," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 1, pp. 53–81, Mar. 2021.

[66] F. Palsson, J. R. Sveinsson, J. A. Benediktsson, and H. Aanaes, "Classification of pansharpened urban satellite images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 5, no. 1, pp. 281–297, Feb. 2012.

[67] L. Wald, *Data Fusion: Definitions and Architectures: Fusion of Images of Different Spatial Resolutions*. Paris, France: Presses des MINES, 2002.

[68] R. H. Yuhas, A. Goetz, and J. W. Boardman, "Discrimination among semi-arid landscape endmembers using the spectral angle mapper (SAM) algorithm," in *Proc. Annu. Jet Propuls. Lab Airborne Geosci. Workshop*, San Diego, CA, USA, 1992, pp. 1–3.

[69] G. Vivone, M. Dalla Mura, A. Garzelli, and F. Pacifici, "A benchmarking protocol for pansharpening: Dataset, preprocessing, and quality assessment," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 6102–6118, Jun. 2021, doi: 10.1109/JSTARS.2021.3086877.

[70] L. Alparone, B. Aiazzi, S. Baronti, A. Garzelli, F. Nencini, and M. Selva, "Multispectral and panchromatic data fusion assessment without reference," *Photogramm. Eng. Remote Sens.*, vol. 74, no. 2, pp. 193–200, Feb. 2008.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 12:30:08 UTC from IEEE Xplore. Restrictions apply.

[71] P. S. Chavez, Jr., S. C. Sides, and J. A. Anderson, "Comparison of three different methods to merge multiresolution and multispectral data: Landsat TM and SPOT panchromatic," *Photogramm. Eng. Remote Sens.*, vol. 57, no. 3, pp. 295–303, 1991.

[72] A. Garzelli, F. Nencini, and L. Capobianco, "Optimal MMSE pan sharpening of very high resolution multispectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 1, pp. 228–236, Jan. 2008.

[73] G. Khademi and H. Ghassemian, "Incorporating an adaptive image prior model into Bayesian fusion of multispectral and panchromatic images," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 6, pp. 917–921, June 2018.



Lihui Chen received the B.Sc. degree in electronics and information engineering from Sichuan University, Chengdu, China, in 2018, where he is currently pursuing the Ph.D. degree in information and communication engineering with the College of Electronics and Information Engineering.

He is interested in image processing, remote sensing, and deep learning.



Zhibing Lai received the B.Sc. degree in electronics and information engineering from Sichuan University, Chengdu, China, in 2019, where he is currently pursuing the M.Sc. degree in pattern recognition with the College of Electronics and Information Engineering.

He is interested in remote sensing image fusion and deep learning.



Gemine Vivone (Senior Member, IEEE) received the B.Sc. *(summa cum laude)*, M.Sc. *(summa cum laude)*, and Ph.D. degrees in information engineering from the University of Salerno, Fisciano, Salerno, Italy, in 2008, 2011, and 2014, respectively.

In 2014, he joined the North Atlantic Treaty Organization (NATO), Science and Technology Organization (STO), Centre for Maritime Research and Experimentation (CMRE), La Spezia, Italy, as a Scientist. In 2019, he was an Assistant Professor with the University of Salerno, Fisciano, Italy. He was a Visiting Professor with the Grenoble Institute of Technology (INPG), Grenoble, France. He is currently a Researcher with the National Research Council, Rome, Italy. His main research interests focus on statistical signal processing, detection of remotely sensed images, data fusion, and tracking algorithms.

Dr. Vivone received the Symposium Best Paper Award at the IEEE International Geoscience and Remote Sensing Symposium (IGARSS) in 2015, the Best Reviewer Award of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING in 2017, and the IEEE-Geoscience and Remote Sensing Society (GRSS) Early Career Award in 2021. He is also the Co-Chair of the IEEE Image Analysis and Data Fusion Technical Committee. He is also an Associate Editor of the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS (GRSL) and an Editorial Board Member of *Remote Sensing* (MDPI), *Sensors* (MDPI), and *Encyclopedia* (MDPI). He served as a guest associate editor for several special issues.



Gwanggil Jeon (Senior Member, IEEE) received the B.S., M.S., and Ph.D. *(summa cum laude)* degrees from the Department of Electronics and Computer Engineering, Hanyang University, Seoul, South Korea, in 2003, 2005, and 2008, respectively.

From September 2009 to August 2011, he was with the School of Information Technology and Engineering, University of Ottawa, Ottawa, ON, Canada, as a Post-Doctoral Fellow. From September 2011 to February 2012, he was with the Graduate School of Science and Technology, Niigata University, Niigata, Japan, as an Assistant Professor. From December 2014 to February 2015 and June 2015 to July 2015, he was a Visiting Scholar with the Centre de Mathématiques et Leurs Applications (CMLA), École Normale Supérieure Paris-Saclay (ENS), Cachan, France. From 2019 to 2020, he was a prestigious Visiting Professor with the Dipartimento di Informatica, Università Degli Studi di Milano Statale, Milan, Italy. He is currently a Full Professor with Incheon National University, Incheon, South Korea. He was a Visiting Professor with Sichuan University, Chengdu, China; Universitat Pompeu Fabra, Barcelona, Spain; Xinjiang University, Urumqi, China; the King Mongkut's Institute of Technology Ladkrabang, Bangkok, Thailand; and the University of Burgundy, Dijon, France.

Dr. Jeon was a recipient of the IEEE Chester Sall Award in 2007, the Electronics and Telecommunications Research Institute (ETRI) Journal Paper Award in 2008, and the Industry-Academic Merit Award by the Ministry of Small and Medium Business (SMEs) and Startups of Korea Minister in 2020. He is also an Associate Editor of *Sustainable Cities and Society*, IEEE ACCESS, *Real-Time Image Processing*, *Journal of Systems Architecture*, and *Remote Sensing* (MDPI).



Jocelyn Chanussot (Fellow, IEEE) received the M.Sc. degree in electrical engineering from the Grenoble Institute of Technology (Grenoble INP), Grenoble, France, in 1995, and the Ph.D. degree from the Université de Savoie, Annecy, France, in 1998.

Since 1999, he has been with the GIPSA-Lab, University Grenoble Alpes, CNRS, Grenoble INP, Grenoble, where he is currently a Professor of signal and image processing. He has been a Visiting Scholar with Stanford University, Stanford, CA, USA; the KTH Royal Institute of Technology, Stockholm, Sweden; and the National University of Singapore, Singapore. Since 2013, he has been an Adjunct Professor with the University of Iceland, Reykjavik, Iceland. From 2015 to 2017, he was a Visiting Professor with the University of California at Los Angeles (UCLA), Los Angeles, CA, USA. He also holds the AXA Chair in remote sensing and is an Adjunct Professor with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China. His research interests include image analysis, hyperspectral remote sensing, data fusion, machine learning, and artificial intelligence.

Dr. Chanussot was a member of the Machine Learning for Signal Processing Technical Committee of the IEEE Signal Processing Society from 2006 to 2008, and the Program Chair of the IEEE International Workshop on Machine Learning for Signal Processing in 2009. He is a member of the Institut Universitaire de France from 2012 to 2017 and has been a Highly Cited Researcher (Clarivate Analytics/Thomson Reuters) since 2018. He was the founding President of the IEEE Geoscience and Remote Sensing French Chapter from 2007 to 2010, which received the 2010 IEEE Geoscience and Remote Sensing Society (GRS-S) Chapter Excellence Award. He has received multiple outstanding paper awards. From 2017 to 2019, he was the Vice-President of the IEEE Geoscience and Remote Sensing Society, in charge of meetings and symposia. He was the General Chair of the First IEEE Geoscience and Remote Sensing Society (GRSS) Workshop on Hyperspectral Image and Signal Processing, Evolution in Remote Sensing (WHISPERS). He was the Chair (from 2009 to 2011) and the Co-Chair of the Geoscience and Remote Sensing (GRS) Data Fusion Technical Committee from 2005 to 2008. He was the Editor-in-Chief of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING from 2011 to 2015. In 2014, he served as a Guest Editor for the *IEEE Signal Processing Magazine*. He is also an Associate Editor of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, the IEEE TRANSACTIONS ON IMAGE PROCESSING, and the PROCEEDINGS OF THE IEEE.



Xiaomin Yang (Member, IEEE) received the B.S. and Ph.D. degrees in communication and information system from Sichuan University, Chengdu, China, in 2002 and 2007, respectively.

She was a Post-Doctoral Researcher with The University of Adelaide, Adelaide, SA, Australia, for one year. She is currently a Professor with the College of Electronics and Information Engineering, Sichuan University. Her research interests are image processing and pattern recognition.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 12:30:08 UTC from IEEE Xplore. Restrictions apply.